# GREEN CHEMISTRY LAB MANUAL WITH A FOCUS ON BUILDING EXCEL SKILLS

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# Green Chemistry Lab Manual with a focus on building Excel skills

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# TABLE OF CONTENTS

# LICENSING

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# Preface

## Preface

### The Creation of this Open Education Resource (OER)

In 2023, the Illinois State Legislature made available $3 million in funding dedicated to Open Education Resource creation. This money was disbursed to academic institutions throughout the state via a 2-year competitive grant (2024-2026) that was overseen by the Illinois Secretary of State/Illinois State Library and facilitated by the Consortium of Academic and Research Libraries in Illinois (CARLI). This textbook and ancillary materials were part of Trinity Christian College's grant submission entitled *OER CARES: Cultivating Academic Resources to Equip Students*. From the grant, four separate projects were funded, including this one created for CHEM 103: Fundamentals of Chemistry I.

The grant's primary purpose is to develop Open Educational Resources (OER) throughout the state of Illinois to reduce textbook costs for students and encourage student success. Developed OER course materials will be released under a license that permits their free use, reuse, modification, and sharing with others.

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### Editors - Sarah Hoeksema, Dr. Kyle McCarrell

### Contributors

Contributors to the development of this OER textbook/ancillary material include educators, librarians, and staff from Trinity Christian College.

### Illinois Secretary of State/Illinois State Library/CARLI Project Staff

The State of Illinois OER grant project would not have been possible without the support of the Illinois State Library and CARLI staff. Topical experts shared their knowledge on open educational resources, accessibility, author rights, instructional design, and publishing.

- Alexi Giannoulias, Illinois Secretary of State
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- Anne Craig, CARLI Senior Director
- Michele Leigh, Project Director, CARLI Senior Coordinator for Open Illinois
- Elizabeth Clarage, CARLI Director of Collections Services and OER Committee Liaison
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### **Peer Review**

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# 1: FIRST DAY OF LAB AND MEASUREMENTS AND ERRORS

**Lab 1: First Day of Lab / Measurements and Errors**

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**FIRST DAY OF LAB**

**Partner Work**: Objectives of Course

Discuss with your lab partner:

What are three things that you would like to get out of this lab course? If you don't want anything out of this lab course, what are three reasons you are here?

| Student Objectives |
| --- |
| 1. |
| 2. |
| 3. |

Looking around the lab, what are you curious about? Is there anything that you are nervous about? Discuss these questions / fears / curiosities with your partner and list some of them below:

| Questions / Fears / Curiosities |
| --- |
| 1. |
| 2. |
| 3. |
| 4. |
| 5. |

**Search and define these common laboratory tools and techniques:**

G1. Taring a scale: ______________________________________________________________________

G2. Meniscus: __________________________________________________________________________

G3. Erlenmeyer flask (draw the shape):

G4. Graduated cylinder (draw the shape):

G5: Volumetric flask (draw the shape):

G6. Scoopula (draw the shape):

**Lab Map:** Draw a map in the space below, indicating the location of the following items. You probably need to get up and explore, perhaps even into the hallway.

1. All Fire Extinguisher(s)
2. Safety Shower(s)
3. All Exits
4. Fume Hood(s)
5. First Aid Kit(s)
6. Chemical Spill Kit
7. All Eye Wash Stations
8. SDS (Safety Data Sheets)
9. Your Workstation
10. Chemical Storage Areas
11. All Balances
12. Glass Waste Container
13. Any Instrumentation

**Partner Work**: Lab rules - why do you think we have the following rules (the "Pros")? Are there any reasons you might not follow them (the "Cons")?

| Rule | Pros | Cons |
|---|---|---|
| Wear approved safety goggles at all times | | |
| No shorts, open-toed shoes, or exposed midriffs | | |
| No food and drink in lab | | |
| Know the location of all eyewash stations, safety showers, and exits | | |
| Tie back long hair and lose clothing | | |
| Do not use broken or cracked glassware | | |
| Never taste chemicals | | |
| Smell only by wafting | | |
| Keep a clean workspace; clean up before leaving | | |
| Report all accidents in the lab, even if you think they are minor | | |
| All book bags and coats must be stored in the designated area | | |
| Only dispose of waste in designated containers | | |

**Introduction to Measurements and Errors**

One of the most important skills in science is being able to measure accurately and precisely. Early in the curriculum, the exact tools you must use are often given. Eventually you will choose your own. This lab aims to introduce you to common laboratory glassware and evaluate which types are best for the job. We will also introduce you to properly using electronic laboratory scales.

There are two types of variations present in all measurement: accuracy and precision. Accuracy is how close a measurement is to the actual value. For example, if a rock has an actual mass of 3.142 grams and our balance reads the mass as 3.141 grams, that is a fairly accurate measurement. If it reads 3.124, that is much less accurate. For determining accuracy, you will calculate the mean (average) of several measured values and compare them to the actual values. This is calculated as , where is the sum of all values and *n* is the number of values:

$$\bar{x} = \frac{\sum x_i}{n}$$

Precision is not about how close a measurement is to the true value, but instead how close repeated measurements of the same value are to each other when repeated. If you take a reading and there is no variation, you are probably not measuring as precisely as you could be. We will look at precision by calculating a standard deviation (the symbol s or sometimes st. dev.) according to the following equation

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{(n-1)}}$$

where n is the number of repeats, $x_i$ is the $i^{th}$ repeat measurement, is the mean of all repeats and the summation (∑) is from i = 1 to n. Microsoft Excel, available on all school computers, has prebuilt formulas for calculating standard deviation that we will use in lab.

Balances are often more precise than volume measurements, as volume can change with temperature. To determine the accuracy of your volumes measured, you will measure their mass and use a known density to calculate actual volumes. It is important to understand that all measurements (mass included) have error. But the error in our masses (if obtained properly) for this experiment are much less than the error in volumes. So, except for Part A of this lab, we will assume that there are no errors in our mass measurements.

Since water is safe and readily available, we will use mass of water to calculate the true volumes of water. The density of water is variable with temperature, so we will determine that together as a class first.

**PROCEDURE**

**PART A (Error in Mass Measurements using an analytical balance)**

1. Since analytical balances can even detect the mass of fingerprints, use a Kimwipe® to obtain three quarters. Use a Kimwipe® when handling the quarters in this experiment.

**2. Determining measurement error of an analytical balance.**

Use the analytical balance and one of the three quarters. Zero the balance with the lid closed, then place a quarter on the pan and close all doors, obtain its mass, and record your reading. Remove the quarter, close the lid, zero the balance again, put the same quarter back on the balance, close doors, and record a second mass. Repeat this process two more times, obtaining a total of four mass measurements for the quarter. After lab period, create a table in Excel similar to the table shown below and calculate the mean using the = AVERAGE() function in Excel and the standard deviation using the =STDEV.S() function in Excel. It is essential you use the correct standard deviation formula (STDEV.S rather than STDEV.P), as the calculation for the standard deviation for samples is slightly different, especially with low numbers of samples.

| | | Mass (g) | | |
|---|---|---|---|---|
| Coin 1 | rep 1 | | | |
| | rep 2 | | | |
| | rep 3 | | Mean | St. dev. |
| | rep 4 | | | |

**3. Determining the error in the mass between several quarters.**

Zero the balance with nothing on the pan (close doors as described in PartA.2). Measure the mass of the second quarter and record it on your sheet for quarter 2. Remove the quarter. Zero the balance and measure the mass of the third quarter. After lab period, add the following below the table you just created, and calculate the mean and standard deviation using the first value for Coin 1, and the values you just measured.

| | | Mean | St. dev. |
|---|---|---|---|
| Coin 2 | | | |
| Coin 3 | | | |

4. Leave the quarters at the front of the lab.

**PART B (Determining Density of Deionized Water)**

Using volumetric flasks, we will determine the density of water at room temperature.

1. Measure and record the temperature of the Deionized Water used in this experiment.

2. Each person will be assigned and given a volumetric flask from the cart of volume 5, 10, 25, 50, or 100 mL.

First measure the mass of the clean and dry volumetric flask on a top loading balance and record your name (column A), the flask volume (column B), and the flask mass (column C) on the linked Excel spreadsheet under the sheet "Water Density" (sheet names are on the bottom of the window). It may be convenient to display the Excel sheet side by side with this document. Ask your instructor if you don't know how to do this. Fill the volumetric flask properly using the meniscus line with deionized water, using a narrow tip Beral plastic pipet to obtain the exact amount very accurately. Measure its mass using the same top loading balance and record the mass on the linked Excel spreadsheet (column D). Create a formula in Excel for the mass of water by subtracting the mass of the flask from the mass of the flask with water (ie "=D8-C8") in column E.

3. As you enter When everyone has entered their data into the spreadsheet, Excel will give line of best fit (regression) data. Make sure to wait until all data is entered to pull density data for your calculations.

**PART C (Determining Error in Volume Measurements)**

1. Follow the directions given for your assigned glassware carefully. The maximum volume of your container is *not* the volume you are trying to measure. For all glassware, record name and mass data for this section in the linked Excel spreadsheet by ***changing the sheet to "Volume Errors".***

**Glassware 1**: Obtain and record mass of a dry, empty 50 mL beaker. Using the markings on the beaker, fill it with 30 mL of deionized water as accurately as possible. Obtain mass of beaker + water and record.

**Glassware 2**: Obtain mass of dry, empty 50 mL graduated cylinder; fill it accurately with 25 mL of deionized water. Obtain mass of the graduated cylinder + water and record.

**Glassware 3**: Using your 10 mL graduated cylinder, obtain mass of dry, empty graduated cylinder; fill it as accurately as you can with 5 mL of deionized water. Obtain mass of the graduated cylinder + water and record.

**Glassware 4**: Obtain mass of dry, empty 50 or 150 mL beaker and record on your report sheet. Using a buret that has been set up by your instructor, fill the beaker as accurately as you can with exactly 10 mL of deionized water that you dispense from the buret. Record the initial buret reading and also record the target buret reading you are aiming for to dispense 10 mL of water. Obtain mass of beaker + water. Calculate and record mass of water on your report sheet.

2. ***Download a copy of the linked Excel spreadsheet and save your own local copy. Open this copy in Excel locally.*** Calculate the mass of water in each container in column F row 8 using a formula to subtract the mass of the container from the mass of the container and water combined.

3. Click in cell F8 so it is highlighted. Hover over the bottom right corner so that your cursor becomes a large cross. Click and drag to the bottom of the range of data to apply this formula in each row.

4. Convert the mass of your measurement into mL of deionized water by creating a formula in column G. Density is mass divided by volume or mass per volume (e.g. grams/mL). Thus, dividing a mass of water by water's density converts grams of water into mL of water.

5. Apply this formula to all rows using the same process you used in step 3.

**PART D (Pipetting Technique and Errors)**

1. For this section *switch the sheet of your local copy to "Pipette Errors"* Obtain a 1.0 mL pipet and a blue tip. Adjust the pipet to as close to 1.0 mL as possible and affix the blue tip. To make sure the pipet is working correctly, gently draw up 1.0 mL of deionized water as demonstrated by your instructor. Then holding the pipet vertically, raise the tip out of the liquid and hold it still for about 15 seconds. If the pipet does not drip, we know that there are no glaring problems with the pipet.

2. Obtain a small beaker, place it on a top loading balance and tare it so that the balance reads zero. Now pipet a 1.0 mL quantity of deionized water into the tared beaker using the 1.0 mL pipet. Record the mass on your spreadsheet.

3. Do **not** tare the balance between readings. Add another 1.0 mL amount with the pipet and record the new mass. Repeat this three more times for a total of five pipettings into the beaker. Make sure you record the new mass after each 1.0 mL pipetting.

4. Calculate the percent error in Excel. Show these values to your instructor to obtain instructor's initials. If the measurement is not accurate enough, then you will need to try again.

**Final Questions: To be answered and submitted digitally with your Excel file. Show all calculations, always including appropriate units.**

**Part A**

1. Submit an Excel file with calculations as described in Part A. Make sure all data is clearly labeled.

2. What did you learn from the four measurements of the mass of the same quarter?

3. Compare the measurements of the single coin vs. the measurements of three different coins. Which set of measurements shows more variability? How did you determine this? What accounts for the large variability in the least precise set of measurements?

**Part B**

4. What is the density of water at our room temperature as determined by the whole class? The actual density of deionized water is 0.9970 g/mL at 25.0 °C and 0.9982 g/mL at 20.0 °C. Using interpolation (either look it up or ask your professor if you do not know what this is) between these two values, calculate what the density would be at our room temperature assuming a linear relationship. Use % difference to determine how close this value is to the class's value.

% difference = * 100

**Part C**

5. Calculate the mean and standard deviation for each type of glassware using your local copy of the Excel file and Excel formulas.

6. Calculate the % difference between measured value (mean) and calculated value (target volume) for each of the four types of glassware (see equation in Q5). Construct a table that includes the type of glassware with corresponding percent errors and standard deviations. Then answer these questions:

a. Which appears to be the most accurate? Explain your answer.

b. Which appears to be the most precise? (Remember these are not the same thing) Explain your answer.

**Part D**

7. Think through your use of the pipet. In what ways can one slip up and not deliver the correct volume if care is not used? Describe as many as you can think of.

8. Explain how you would perform each of the following including the type of measuring tool (glassware or pipet) you would use. Pay attention to these descriptions so that you can properly balance between accuracy, precision, & ease.

a) As accurately as possible deliver 12.35 mL of solution A into a reaction mixture.

b) Obtain about 20 mL of a solution that you will use later in the experiment.

c) Obtain 21.5 mL of a solution and add all of it to 5.00 g of a solid that has been put into a beaker.

d) Measure the amount of liquid you added to react with a solution as precisely as possible.

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# 2: DILUTIONS AND DENSITIES

## LAB 02 – DILUTIONS AND DENSITIES

## PART A: DILUTIONS

**Background:**

Last week, we found that it is often easier to measure mass precisely than volume. However, when very small masses are needed in relatively large volumes, we often cannot measure small enough masses without using very large volumes. For example, we often measure metals in solutions in units in parts per million, meaning that if we wanted 1.00 grams of zinc in a solution, we would need to have a million grams of solution. For water, this would be about 1000 L. It would be very wasteful to prepare solutions this way, so instead solutions are prepared and then diluted to various concentrations as needed. We will be working through dimensional analysis for these dilutions to discover the most important equation for dilutions, and probably the most common equation used by research chemists and cellular biologists. We will be using a compound that absorbs light linearly with concentration, which allows us to see quickly if the dilution went well.

The dilution in this lab is a 1000-fold dilution and cannot be done accurately in a single step. A more accurate way is to dilute your starting solution part way and then dilute this new solution again. Several of these steps can be used to make extreme dilutions from a starting solution and still be very accurate. This method is called a serial dilution because each new dilution is subsequently diluted again, making increasingly more dilute solutions.

**Partner Work:**

1. Using a balance and weighing paper, obtain about 0.1 (±0.005) g of the reddish colored powder (fluorescein) from the vial next to the balance. Record actual mass with all available significant figures. Be as careful and clean as possible. Immediately add the powder to a 50 mL beaker. Rinse all powder into the beaker with a small amount of DI water from a water bottle. Add about 10 mL of 0.2 M $Na_2CO_3$ base to this solid and dissolve the solid by carefully stirring with a glass stirring rod.

2. Once dissolved, quantitatively transfer (transfer it all by rinsing with multiple small volumes) this solution into a clean 25 mL volumetric flask by successively washing the beaker with small amounts (3-5 mL) of DI water, pouring each rinse into the flask being sure to stay under the 25 mL mark. Use a beral pipet to accurately reach the 25 mL mark. If you go over the 25 mL mark, you must do step 1 and 2 over. Seal with Parafilm® and mix well by inverting and swirling several times.

**Questions:**

1. What is the actual mass, including all significant figures and units of your fluorescein?

2. What is the actual volume, including an appropriate number of significant figures and units? This should include the decimal place indicated by the tolerance of the flask – for example a 25 mL volumetric flask with a tolerance of ± 0.1 mL should be reported as 25.0 mL, while a 25 mL ± 0.01 mL volumetric flask is reported as 25.00 mL.

3. Do you think that your mass reading or your volume reading introduces a larger degree of uncertainty (less precision)? Why?

4. Concentration units are often given in a mass per volume or an amount per volume. What is the concentration of the fluorescein solution you made in these steps? Included the appropriate number of significant figures and units.

**Partner Work:**

3. Be very accurate in this series of steps. Note: 5.0 mL of a solution plus enough DI water added to reach a total of 50.0 mL yields a10-fold dilution because the volume has increased 10-fold while the amount in the 5.0 mL hasn't changed.

a) Obtain 5.0 mL of your solution using your 10 mL graduated cylinder and quantitatively transfer it to your 50 mL graduated cylinder with several small washes of DI water, bringing the total volume up to 50.0 mL. Seal your 50 mL graduated cylinder with parafilm and invert several times to mix. You now have made a 10-fold dilution of your original solution. Place this solution in a clean beaker.

b) Repeat step 3. a) with this exception, measure 5.0 mL of your 10-fold diluted solution instead of your original solution. Adding DI water to make 50.0 mL will give a 100-fold dilution of your original (10-fold times 10-fold). ***This should be visibly more dilute than the previous solution.***

c) Repeat step 3 a) a third time, using 5.0 mL of the 100-fold diluted solution. After the final mixing, you have a 1000-fold diluted solution. As can be seen, this sequence of three steps is called a serial dilution since each solution is further diluted by the next consecutive step in the sequence.

**Questions:**

5. What is mass of fluorescein represented by the 5.0 mL sample in step 3a? Use the 5.0 mL volume, your concentration from question 4, and dimensional analysis. Make sure to include the correct number of significant figures and units.

6. After bringing the volume up to 50.0 mL in step 3a, does the mass of fluorescein present change? Why or why not?

7. Using the same concentration units you used in question 4, what is the new concentration of fluorescein at the end of step 3a? At the end of step 3b? At the end of step 3c? Make sure to use an appropriate number of significant figures and units for each answer.

8. Notice the general form of your calculations. As you dilute from initial concentration, $C_1$, with a volume, $V_1$, to a new volume, $V_2$, you arrive at a new concentration, $C_2$. The calculation you performed by dimensional analysis is $C_1*V_1/V_2 = C_2$. ***The dimensional analysis always works out if the units for concentration and volume are the same for each value.*** What would the final concentration units need to be if your initial concentration was in mg / dL?

**Partner Work:**

4. To check your work, obtain the absorbance of the final solution (the most diluted made in 3c) with the LabQuest and SpectroVis instrument as follows:

a) Use the USB cable to connect the SpectroVis to the LabQuest and then turn on the LabQuest with the power button in the top-left corner. Obtain a plastic cuvette (do not touch its clear sides, only its frosted sides), fill about ¾ full with DI water, and carefully place in the SpectroVis so that the clear sides are lined up with the light and detector. Calibrate the spec by selecting the following menus/buttons with the stylus (NEVER A PEN OR PENCIL) (Sensors / Calibrate / USB:Spectrometer / {wait for the warm up period} / Finish Calibration / OK).

b) Remove and empty the cuvette, fill about ¾ full with your 1000-fold dilution (3c), and place in the spec. Start data collection by tapping the green arrow button in lower left corner of screen (select discard if asked) and then the red stop button (located in the same spot because green arrow turns into red square).

c) Select the wavelength with the highest absorbance (using the left and right arrow keys if needed). If it is not close to 490 nm, consult your instructor. Record this wavelength and the absorbance value (Abs) given. Show these values and the spectrum to your instructor for validation.

5. Label a white piece of paper with numbers 1-6, spaced enough that each number could have a cuvette placed on top. Place the cuvette from step 5 over a piece of paper labeled "1". Using a calibrated micropipette, add the following amounts of deionized water and your most dilute fluorescein solution (3c) to create cuvettes 2-6.

| **Cuvette** | **2** | **3** | **4** | **5** | **6** |
|---|---|---|---|---|---|
| Water (mL) | 0.5 | 1.0 | 1.5 | 2.0 | 2.5 |
| Fluorescein (mL) | 2.5 | 2.0 | 1.5 | 1.0 | 0.5 |

**Questions:**

9. Based on your LabQuest spectrum, why is it important to specify a wavelength when working with light absorption?

10. When looking at your solution from 3c in the cylinder and the solution of the same concentration in cuvette 1, does one appear to absorb more light than the other? If they are the same concentration, why do you think this is?

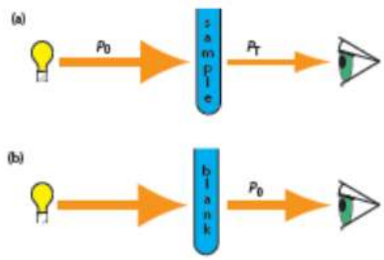


FIGURE 1

**1a)** Incident power ($P_0$) is diminished to transmitted power ($P_T$) as it passes through a sample to a detector. **1b)** Incident power and transmitted power are the same (P0) when only a blank sample is present.

From: Analytical Chemistry 2.1 by David Harvey (summer 2016) via (CC BY-NC-SA 4.0)

**Background:**

Concentrations are difficult to measure directly. You could, in theory, evaporate all the solvent off and measure the remaining mass. That would work if you only had one dissolved component in a mixture. In most cases, we take advantage of relative consistency in absorptivity for compounds to establish standardized controls that relate concentration and absorbance.

Absorbance is a calculated value based on what we can actually measure – the intensity or power of light. **Figure 1** shows how the power of light detected changes as the sample absorbs some portion of light.

**Questions:**

11. Look at the 6 cuvettes you created in step 5. Which is most concentrated? How does the color change with concentration?

12. As you diluted your fluorescein solutions, what happened to the amount of transmitted light through the solution? Was transmission higher or lower at high concentrations?

13. How would you express the amount of light transmitted as a percent of the maximum possible light transmitted, using the variables from **Figure 1**?

14. ***Download a copy of the*** *linked Excel spreadsheet* ***and save your own local copy. Open this copy in Excel locally.*** Switch the sheet in your local, downloaded copy of the linked Excel spreadsheet to "Beer's Law". A table has been created and partially completed. When you have data that varies predictably, Excel can auto-fill the data. Click and drag from Cell B4 to Cell C5 to highlight a 2x2 block of cells. Click on the bottom right of the block (the icon should be an all-black cross) and drag to column G. Verify that the amounts indicated match those from step 5.

15. Enter the concentration of fluorescein you obtained at the end of step 3c (see question 8) into the cell B6. Notice that the cell to the right automatically updated, as it was filled in with a formula. Click on the cell (C6) to view the formula. Near the top of the window, the function should be visible as =$B$6*C5/(C4+C5). Cell B6 is the concentration of the fluorescein solutions you are using for ***every one*** of the dilutions ($C_1$), C5 is the volume of the fluorescein solution ($V_1$) you are diluting to the final volume, $V_2$, which is the sum of the volume of fluorescein and the volume of water used to dilute it.

16. Click in the bottom right of cell C6 and drag to G6. What is the formula now listed in cell G6?

17. What did placing $ in front of the row and column labels for B6 do when copying the formula to adjacent cells?

**Partner Work:**

5. On the graph screen of the LabQuest, make sure the wavelength of maximum absorbance is still selected. Return to the meter screen using the meter icon in the top banner, and set the units for the spectrophotomer to %T by clicking on the red banner. Select "Change Units" and "% Transmittance".

6. Measure the % transmittance for each cuvette 1-6, carefully returning each after you are finished. Record these values in your local copy of the linked Excel spreadsheet.

7. Absorbance is a calculated value defined as the negative log of transmittance (what isn't transmitted is absorbed). Since transmittance is often reported as a percent, this needs to be corrected in the calculation by dividing % Transmittance by 100 to express as a decimal. This calculation is completed in cell B8. Drag it over to extend it through G8.

**Questions:**

18. As you filled out the spreadsheet, a graph relating %T vs. Concentration should have automatically filled in your data. Does your measured relationship between % transmittance and concentration match what you observed with your eye in question 17? Is it a linear relationship?

19. Create your own graph relating Absorbance vs. Concentration:

a. Navigate to the insert menu of Excel. Click on the charts submenu :


b. Choose "Scatter", the icon on the top left

c. Click and drag the blank graph window so that it is below the %T vs. Concentration graph.

d. Excel should default to "Chart Design" when the chart is selected. Choose "Select Data" toward the top left of the menu ribbon or right click on the chart and choose "Select Data".

e. Under "Legend Entries (Series)" on the left, click the "Add" button

f. Type "Absorbance vs. Concentration" under "Series name:"

g. Click the arrow next to "Series X values:" Click and drag over your concentration values, which will be plotted against the x-axis. Hit enter or return to validate the selection.

h. Click the arrow next to "Series Y values:". Click and drag over your Absorbance values, which will be plotted against the y-axis. Hit enter or return to validate the selection. Hit OK twice to close the remaining windows

20. Is your plot of Absorbance vs concentration linear?

**Background:**

You should have a relatively straight line for absorbance, but not for % transmittance. Earlier you saw that the amount of light absorbed is related to 1) the concentration of the sample, 2) the length of the path through the sample that the light travels (Q16), and 3) the wavelength of light being used. But a fourth component is the intensity of the light itself. As the light travels through the sample it is being absorbed, which in turn reduces the amount of light that can be absorbed as the light continues to travel through more of the sample. This constant light loss increases the complexity of light absorption, and we must use calculus to get an accurate equation describing the absorbing process. Using calculus, we find that the integral of dP/P must be taken, which introduces a log function resulting in the following equation:

$$-\log\left(\frac{P}{P_0}\right) = \varepsilon * \ell * C$$

where e (the molar absorptivity) is a constant for each wavelength for each compound, $\ell$ is the path length that the light travels through the sample (our cuvettes have the path length of 1 cm), and C is the concentration of the compound that absorbs the light. This $-\log(P/P_0)$, is also called the Absorbance (which is dependent upon the wavelength of light), and through substitution we can write

$$Abs = \varepsilon * \ell * C$$

This is the Beer-Lambert Law, and is a powerful way to measure concentrations of solutions.

**Questions:**

21. Click on any of the data points displayed on the absorbance graph to select the data series, then right click and select "Add Trendline". An options window or menu should appear, with a "Linear" trendline set as the default. Scroll down in this menu to check "Display Equation on chart" and "Display R-squared value on the chart". The equation will be in the form y=mx+b, where y is the Absorbance and x is the concentration. The b, the x-intercept, should be very close to zero indicating little to no Absorbance for the blank. If the b value is zero, this m value (the slope) should be equal to the e (molar absorptivity) * $\ell$ (path length) from the Beer-Lambert Law. Given the path length is 1 cm, what is the molar absorptivity with units of the fluorescein based on the slope of your trendline?

## PART B: DENSITY OF A LIQUID

**Background**

In the next two parts of lab you will determine the densities of an unknown liquid and a solid. The simple process for this is to measure the mass and volume of the unknown and then calculate the density (ratio of your two measurements). Both measurements need to be accurate to obtain an accurate density!

Because volume measurements are temperature dependent and typically harder to read with precision, we will use a method that imitates a pycnometer. A pycnometer is a special flask with a ground glass top with a small hole in the top. The volume of liquid left in this flask after excess liquid has been pushed out by the stopper is more precise than using your eye to read a meniscus. We will use a simple volumetric flask with a glass stopper to work like a pycnometer for accurate volume measurements, but we will need to calibrate the volumetric flask using the density we calculated earlier.

You may find it convenient to record values on page 5 of this lab, on a separate piece of notebook paper, or in the linked Excel spreadsheet by switching the tab to "Densities". If you use the spreadsheet, your work can be within the cell formulas and does not need to be recorded separately, but only if you use formulas to do the calculations!

**Partner work:**

1. Obtain an unknown liquid from the front of the lab and a clean, dry 10 mL volumetric flask with a glass stopper to serve as our pycnometer. Record the label on your unknown liquid on your report sheet. Record the room temperature to a tenth of a degree. (Assume that your unknown liquid and the DI water are both at this temperature). Obtain and record the mass of the dry stoppered flask as accurately as possible.

2. Fill the flask completely full with your unknown liquid and then stopper your flask, being careful not to trap any air bubbles in the flask. A small amount of liquid should be pushed out of the flask as it is stoppered. While keeping the stopper on the flask, carefully dry off any excess liquid from the outside of the flask and stopper. Obtain the mass of the stoppered flask and liquid. Record on report sheet.

3. Empty the liquid back into its original vial and return the capped vial of unknown to the front of the lab. Rinse the flask and stopper with about 5 mL of acetone, and finally rinse twice with DI water. Repeat step B. 2 using DI water instead of the unknown. Obtain and record the mass of the stoppered flask and water.

4. Using the recorded temperature from the report sheet, determine the density of water at this temperature using interpolation if necessary. Record on report sheet.

5. Use the mass of the water and its density at the temperature of the lab to determine the volume of the flask.

6. Finally use these accurate measurements of the mass and volume of the unknown liquid to calculate its density.

**PART B Density of a Liquid Show ALL calculations** Unknown's label ______________

Mass of dry flask and stopper (measure) ________________

Mass of flask, stopper and Unknown liquid (measure) ________________

Mass of Unknown liquid (calculate) ________________

Mass of flask, stopper and DI water (measure) ________________

Mass of DI water (Calculate) ________________

Density of water at ________°C (room's temp) (from part A) ________________

Volume of flask with stopper inserted (calculate) ________________

Density of unknown liquid (calculate) ________________

**Questions:**

22. What was the purpose of the water?

23. Why is the stopper so important? Can't you just fill the flask up to the brim? Why not use the line on the volumetric flask in a normal way?

## PART C: DENSITY OF A SOLID

**Background:**

Measuring the volume of a solid may seem difficult at first. However, you will use a displacement method where both the solid and water (the density of which you can determine easily) combine to fill a precise volume. If you know the mass of the metal, you can determine the mass of the water in the container. The mass of water in the container and its density are used to determine the volume of the water. If you know the total volume of the container and the volume of the water covering the metal, then how would you calculate the volume of the solid?

You may again find it convenient to record values on page 6 of this lab, on a separate piece of notebook paper, or in the linked Excel spreadsheet by switching the tab to "Densities". If you use the spreadsheet, your work can be within the cell formulas and does not need to be recorded separately.

**PART C Determining the density of a solid by displacement**

1. Dry your flask and stopper by rinsing with a small amount of acetone (acetone evaporates much faster than water). Copy the mass of the empty flask and stopper from Part B above.

2. Obtain an unknown solid and record its label on your report sheet. Add enough of the solid to fill the flask at least half full or until you use all your solid, whichever occurs first. Obtain and record the mass of the flask, stopper and solid.

3. Add enough water so that the solid is completely covered with water. Swirl the mixture to remove all air bubbles clinging to the solid particles. You may need to gently shake the flask to dislodge the last of the air bubbles. Now, completely fill the flask with DI water and stopper it so as to remove all air and excess water. Carefully dry the outside of the stoppered flask and obtain the mass of the stoppered flask + metal + water. Record on your report.

4 From these observed values calculate volume and mass of the solid and, finally, calculate its density. Record on report sheet.

5. Decant the water from the flask, keeping all the metal. Pour the wet metal onto 3-4 layers of paper towel to soak up excess water from the metal. Return the metal to the proper vial and put in the oven at the back of the lab without the cap on. Place the cap on top of the oven. When dried, your instructor will take care of the vials.

**PART C Density of a Solid Show ALL calculations** Unknown solid's label ________________

Mass of dry flask and stopper (from Part B) (copy) ________________

Mass of flask, stopper and Unknown solid (measure) ________________

Mass of Unknown solid (measure) ________________

Mass of flask, stopper, Unknown and DI water (measure) ________________

Mass of DI water (calculate) ________________

Density of water at ________°C (room's temp) (copy) ________________

Volume of flask with stopper inserted (see part B) (copy) ________________

Volume of DI water (calculate) ________________

Volume of the solid (calculate) ________________

Density of solid (calculate) ________________

## QUESTIONS:

24. Suppose that a few small air bubbles were trapped in your flask. What effect would this have on your final estimate of the solid's density? Would the calculated density be too low, too high, or not affected? Explain why.

# 3: ATOMS WHERE YOU LIVE- WATER QUALITY BY MOLECULAR AND ATOMIC SPECTROSCOPY

**Lab 3: Atoms where you live: Water Quality by Molecular and Atomic Spectroscopy**

*adapted from J. Chem. Educ., 2006, 83, 250. &*

*Kegley, S.E., et al. Water Treatment: How Can We Purify Our Water? Student Manual. John Wiley & Sons: New York.* ***2000****. p 35—45.*

Water quality is a hugely important part of public health and safety, but how do we monitor it? Today's lab will explore metal ions as part of water quality, though water quality involves many other factors as well. We will specifically focus on measuring $Fe^{2+}$, $Na^{+}$, $Ca^{2+}$, and $Mg^{2+}$ in water. $Ca^{2+}$ and $Mg^{2+}$ are the primary cause of hard water; due to their low solubility, they easily precipitate and cause solid scale to form and can clog up pipes. Other factors to water quality we will not look at today include the pH of water, and the presence of pathogens or toxic metals and compounds.

In parts A-B, we will use a spectrophotometer to measure the amount of $Fe^{2+}$ in solution through absorbance. In order for this to work, we need to have a compound with visible color, so we will react the $Fe^{2+}$ with phenanthroline, which gives a reddish color that changes in intensity depending on the amount of $Fe^{2+}$ present. This will require making a standard curve (part A) so that we can then test our unknowns (part B), and will be a nice review of what we did in lab last week.

Parts C-D will look at atomic emission spectroscopy, where we detect atoms by observing electron movement between shells when given energy. In part C, we will examine hydrogen, the simplest element with only 1 electron, and then carry principles from that over to look at more complicated metals using the Microwave Plasma Atomic Emission Spectrometer (MP-AES) in the instrument room across the hall. This instrument works by converting all atoms to a high-energy plasma state, which excites electrons in each atom to the highest energy shell possible. When the electrons come crashing back down, you get an extremely large sample per atom, allowing you to detect metals that are present even at very low concentrations.

**Section A: Calibration of the Spectrophotometer**

Just like in our previous lab, the spectrophotometer must be calibrated to understand the relationship between absorbed light and iron concentrations. You will begin by running four reference standards with known iron concentrations: 1 ppm, 2 ppm, 3 ppm, and 6 ppm. We also need a blank, a solution with all reagents except the analyte. Water and other reagents absorb a small amount of light, and this blank allows you to measure that amount so that you can subtract it from the light absorbed by the orange iron-phenanthroline compound.

**Procedure:**

1. Obtain the following solutions in separate beakers from your kits. Carefully label these solutions so that you do not get them mixed up. These are the solutions you will use to prepare your tests and Unknown iron samples.

a) About 20 mL of the 0.538 mM iron solution ($Fe^{2+}$).

b) About 25 mL of 0.002 M phenanthroline (Phen).

c) About 15 mL of sodium acetate solution (NaOAc).

d) About 15 mL of hydroxylamine ($NH_2OH$ ).

e) About 20 mL of deionized water (DI)

2. Download a local copy of the linked Excel spreadsheet, enter the concentration of the available iron standard ($C_1$) in ppm in cell B2.

3. For each target concentration ($C_2$), you will make up a 3.0 mL ($V_2$) solution directly into a cuvette. Label a piece of paper blank, 1 ppm, 2 ppm, 3 ppm, and 6 ppm and place cuvettes above each label. Recall the equation you generated by dimensional analysis was $\mathbf{C_1*V_1 = C_2*V_2}$. *You will use this equation often throughout your science coursework and should memorize it.*

4. To each solution you will add:

a. 0.600 mL of 10% sodium acetate buffer solution.

b. 1.500 mL of 1,10-phenanthroline solution

c. 0.300 mL of hydroxylamine solution

d. Enough iron solution to obtain the target concentration.

e. Enough water to reach 3.000 mL total

5. Create a formula in column B6-B9 that calculates the volume of standard ($V_1$) you must add to get 3.0 mL ($V_2$, column C) of the target concentration ($C_2$, column A), given the standard concentration ($C_1$ in B2).

6. Cell G5 is set up with a formula for how much water needs to be added to get the total water volume up to 3 mL based on all the other components. Drag the formula down to fill cells G6-G9.

7. Make up your solutions according to your calculations. Use a clean tip for each reagent. Add the 1,10 phenanthroline last in two measurements of 0.750 mL. After creating each solution, mix by pipetting up and down repeatedly, moving from low concentration to high.

8. Connect a SpectroVis to a LabQuest. Make sure the units are set to “Abs”. If they aren’t, click on the red banner and select “Change Units”.

9. Click on the red banner and select “Calibrate”. Follow the instructions on screen.

10. Click on the red banner again and select “Mode”. Choose “Events with Entry”, and set the wavelength to 562 nm. For the event, give the label “[Fe]” and the unit “ppm”.

11. Use the “play” button to start taking data.

12. Place your 1 ppm sample in the reader, making sure the clear sides line up with the light and arrow icons on the SpectroVis. Hit “Keep” on the bottom of the screen, and enter “1” for the first event. Repeat for 2, 3, and 6 ppm samples.

13. Hit “stop” to complete data collection.

14. Use the “Table” icon to navigate to your data in table form, and record the absorbance data on your spreadsheet in cells H5-H9.

15. Create a scatterplot with Target concentrations (A6-A9) as the x-axis and Absorbance (H6-H9) as the y-axis. Make sure to label axis and the graph clearly. Refer to last week’s labs if you forgot how to do this.

16. Add a linear-trendline for the displayed data, showing the equation and $R^2$ value.

**Section B: Determination of Iron in an Unknown Sample**

Now that we have a calibration relationship between the iron-phenanthroline complex and absorbance, we can measure the absorbance of samples to determine iron (II) concentration.

**Procedure:**

1. Gravity filter by your sample. (Preparing a filter). Retain this filtered sample for part D as well.

2. Obtain a clean, dry cuvette.

3. Using a calibrated pipette, measure out 0.600 mL of your filtered sample, and add it to a cuvette.

4. To each solution add:

a. 0.600 mL of 10% sodium acetate buffer solution.

b. 0.300 mL of hydroxylamine solution

c. 1.500 mL of 1,10-phenanthroline solution

5. Measure 2 more 3.0 mL samples for a total of 3 measurements. Record each in your local copy of the attached spreadsheet.

6. In the column next to these absorbance values (y), calculate concentration (x) using the equation for the trendline you created in Section A (solve for x in terms of y).

7. Remember that the calibration you performed was based on the final concentrations, but your water sample only makes up 0.6 mL of 3.0 mL. Using the equation $C_1*V_1 = C_2*V_2$, find the concentration of iron in your sample before dilution.

8. Extend these calculation for each absorbance value. Calculate the mean and standard deviation for both absorbance and the concentration.

9. There are better ways to use the linear regression data for your trendline to report uncertainty, but the simplest way of reporting this data is as the mean concentration ± the standard deviation of the concentration.

**Section C: AE Analysis of Hydrogen**

The first technique used in this lab analyzed specifically for $Fe^{2+}$. With AE, you can analyze many ions from the same sample solution, measuring many different water quality parameters at once. Calcium and magnesium are the key ions that contribute to the hardness of water, and are the most abundant cations in solution. Sodium is included for later charge balance analysis. The iron analysis will enable you to compare your results from the AE test with those from the SpectroVis measurements. Before we run the atomic emission experiment for more complex atoms, we will look at hydrogen emission spectrum which matches the material The hydrogen emission spectrum will help us to understand what is happening in the microwave-plasma atomic emission spectrophotometer (MP-AES).

As covered in class, atoms, molecules and subatomic particles have energy. If something gains energy, then its total energy content increases. But if something loses energy then its total energy content decreases. You also know that the ground state is a molecule’s lowest possible energy state while all higher energy levels are called excited states. We have also just learned that these are quantum states, which means that only certain levels of energy are allowed. In this lab we will observe several transitions between these energy states and see how the energy gained or lost must match the difference between the energies of two levels within the molecule.

This energy that is gained or lost can be in the form of thermal energy (from collisions between particles or molecular vibrations) or can be in the form of electromagnetic radiation. We see both of these energy changes at the same time in molecular spectroscopy, resulting in the broad absorption spectrum you saw in parts A and B. Atomic spectroscopy differs from molecular spectroscopy because the lines in atomic spectroscopy are much thinner than the wide bands we see in molecular spectroscopy, owing almost entirely to electronic transitions. Because atoms don’t have bonds like molecules, there is no widening in the electronic transitions from the vibrations of the bonds. Every element has electronic transitions in the UV-Vis region, meaning that we can use this method for most elements, without any sensitizing compound (like the phenanthroline from parts A and B), and because the signals are narrow lines rather than broad bands, modern instruments can resolve many different molecules simultaneously. On the other hand, atomic spectroscopy tends to produce atoms or occasionally ions of the same oxidation state for analysis, regardless of their oxidation state in solution.

We will use electricity to excite hydrogen gas (our simplest atom) and observe its visible emission spectra (Balmer lines).

1. Obtain one spectroscope for your group and the LabQuest setup that consists of a LabQuest, SpectroVis, fiber optics cable, and USB cable. Do not crimp the optics cable! Place the black block at the end of the fiber optics cable into the cuvette position properly (the white triangles aligned with each other) or you will not get any spectra on the LabQuest screen. Change the LabQuest units to intensity.

2. With the LabQuest, observe and record the spectra for hydrogen. Record as many lines as you can see. Fill in the charts on your local copy of the linked Excel spreadsheet on the "Hydrogen Emission" sheet under "Balmer lines".

3. Convert the wavelengths observed in nm to wavelength in meters, then to Energy in kJ / mol (E = hc/λ). Constant values are given in the Excel spreadsheet, so if you use absolute references (ie $A$1), you should be able to create one formula and drag it down.

4. Use the next table to calculate the seven lowest energy levels and the highest possible energy level (accomplished by using an arbitrarily large number) for the electron orbits for hydrogen as described by Bohr $\left(E_n = -\frac{Rhc}{n^2}\right)$. Remember n = energy level and R is the Rydberg constant.

5. The numbers directly from the equation are quite small with units for E given in J/electron. Convert these into units of kJ/mol in the next column.

6. The Balmer lines are all from transitions between higher shells and shell 2. Calculate the energy of these transitions using the "Energy of transitions table" on the spreadsheet.

7. The energy of the Balmer lines should closely align with energy of the electronic transitions. Place the wavelength values you detected next to the energy transitions to which they are closest.

**<u>Section D: AE Analysis of Samples for Mg, Ca, Na, and Fe concentrations</u>**

***Goals:*** *(1) To utilize an analytical tool that can selectively measure ions in a mixed solution. (2) To determine the concentration of* $Ca^{2+}$*,* $Mg^{2+}$*,* $Na^{+}$*, and* $Fe^{2+}$ *in your water sample.*

***Discussion:*** The first technique used in this lab analyzed specifically for $Fe^{2+-}$. With AE, you can analyze four (or more) ions from the same sample solution, thus measuring a number of different water quality parameters at once. We will tests for $Ca^{2+}$, $Mg^{2+}$, $Na^{+}$, and $Fe^{2+}$.

***Experimental Steps:***

1. Take the samples across the hall to the AE.

2. Conduct analyses for Mg, Ca, Na, and Fe. Record into your local copy of the linked Excel spreadsheet, including units. Use this data in the determination of total positive charge in your solutions. You will need to perform unit conversions to get from the units generated by our calibrations (parts per million (ppm) or parts per billion (ppb) to Molar units (M), which are moles per liter. These units are similar in form to percent, literally "per hundred". It may be convenient to think of these as similar units – just as a percent can be thought of as the mass present in 100 grams of solution, ppm can be thought of as the mass present in 1 million grams of solution and ppb can be thought of as the mass present in 1 billion grams of solution. From here you can use the molar mass of the analyte (g analyte / mole analyte) and the density of the solution (g solution / mL solution) to convert to Molar (mole analyte / L solution). You may use Excel to do the calculations, or show your work on paper.

$$x\ mass\ percent\ (part\ per\ hundred) = \frac{x\ g\ analyte}{10^2\ g\ solution}$$

$$x\ ppm\ analyte = \frac{x\ g\ analyte}{10^6 g\ solution}$$

$$x\ ppb\ analyte = \frac{x\ g\ analyte}{10^9 g\ solution}$$

**Post-lab questions**

1) Based on the slope of your trendline for iron analysis, what is the molar absorptivity ( ) of the iron solution at 562 nm?

2) Show that a 0.538 mM solution of ammonium iron (II) sulfate ($(NH_4)_2Fe(SO_4)_2(H_2O)_6$) is 30 ppm in iron by converting the units.

3) Why is it important to calibrate a spectrometer? How does the calibration indicate sensitivity? To what part(s) of the Beer-Lambert law do(es) this relate? Why is the buffer important?

4) In your own words, describe how the molecular spectroscopy works. List an advantage and a disadvantage of this technique compared to atomic emission (AE). Which:

· has a higher signal / atom (sensitivity)

· is easier to prepare

· is faster

· is more accessible

· is more error prone

· can determine ion charge / oxidation state?

5) Another version of this lab uses ferrozine to bind iron. If the calibration has been completed for the iron-phenanthroline complex, reading a iron-ferrozine complex gives an absorbance of nearly zero. Why?

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# 4: HARD-SOFT ACID BASE REMEDIATION

**Lab 4: Hard-Soft Acid-Base Remediation**

*Stathatou, P.M., Athanasiou, C.E., Tsezos, M. et al. Commun Earth Environ* ***3,*** *132 (2022).*

*Sharma, H.; Sharma, A.; Sharma, B.; Karna, S. Hindawi International Journal of Analytical Chemistry*. ***2022***, *ID 8520432*.

**Procedure:**

***Extraction of acacia catechu phenols (groups of 2-4)***

Measure out 25 g of Acacia catechu powder into a 100 of 150 mL Erlenmeyer flask. Add 5 g $NaHCO_3$.

**Background:**

Lewis acids and bases can be further classified as “hard” or “soft” acids and bases. Hard acids tend to have very low electronegativities and small electron spheres. Hard bases tend to have very high electronegativities and small, nonpolarizable electrons spheres. Soft acids and bases tend to have moderate electronegativities and more polarizable electron spheres.

**Questions:**

1. Where in the periodic table would you expect to find a hard Lewis acid? Give 2 examples.

2. Where in the periodic table would you expect to find a hard Lewis base? Give 2 examples.

3. Where in the periodic table would you expect to find a soft Lewis acid? Give 2 examples.

4. Where in the periodic table would you expect to find a soft Lewis base? Give 2 examples.

**Background:**

Because orbitals need to overlap constructively to form bonds, the strongest bonds are typically formed between hard acids and hard bases, which have both have smaller orbitals, or between soft acids and soft bases, which both have larger orbitals. Metal ions in the bottom-center to the bottom right of the periodic table tend to be soft acids, with the notable exception of the lanthanides and actinides in the f-block.

**Questions:**

5. Look back at your notes on ionic radii. Why are the lanthanides and actinides hard acids and not soft?

6. Identify the following toxic metals or precipitating anions as hard or soft.

· $U^{+6}$ · $O^{-2}$ · $F^{-}$ · $Hg^{+2}$ · $Pb^{+2}$ · $Cd^{+2}$ · $Sn^{+2}$ · $S^{-2}$

**Background:**

Heavy metals contamination can pose environmental and health hazards. Unlike organic contamination, however metals cannot be further degraded to a less toxic form. Instead, remediation (or treatment in the case of ingestion) often involves chelation (binding with multiple base atoms on the same molecule), adsorption, precipitation, or demobilization, all strategies that involve binding the metals by other compounds to help them be isolated or made less available.

Chelators and adsorbents often will have multiple atoms that serve as Lewis bases that are covalently linked together. For example, the biodegradable chelator NTA (nitriloacetic acid, see right), has 7 atoms that may serve as Lewis bases.

**Questions:**

7. What two types of atoms serve as Lewis bases in NTA?

8. Is NTA a hard base or a soft base?

9. Do you think most chelators, adsorbents, or precipitants for remediation should be hard bases or soft bases? Consider your answer to 6. Why or why not?

10. Metals like $Ca^{+2}$ and $Mg^{+2}$ are essential to all forms of life. Are they hard or soft acids?

11. What dangers can you think of for using a hard base chelator for remediating a heavy metal in a natural system?

12. Which is a harder Lewis base $O^{-2}$ or $S^{-2}$? Why?

13. Iron occurs naturally as $Fe^{+2}$ and $Fe^{+3}$. Which form is a harder Lewis acid? Why?

14. Which is a more likely compound – FeS or $Fe_2S_3$? Why?

15. Which is a more likely compound – $FeCO_3$ or $Fe_2(CO_3)_3$? Why?

**Background:**

In lab today, we will be testing a bevy of chelating agents and adsorbents to see which may be able to remediate heavy metals. We will be using zinc for initial screenings as a model for soft Lewis acid interactions with our remediation candidates, as zinc waste is not toxic. A common method of metals analysis is by colorimetric determination – we did this last week with an iron – ferrozine complex. Sometimes the compounds combined with the metals are toxic, and can create more complex waste for disposal. We are using acacia leaf powder, rich in catechols that create colored complexes with many metals, to determine zinc concentrations without generating hazardous waste. When we add our remediation candidates, only zinc that is unbound (free zinc) can react with the catechols to form a colored solution, so we can tell how much has been "remediated".

**Procedure:**

***Colored Reagent Solution (groups of 2-4):***

1. Filter the acacia powder extract using a Buchner funnel, filter paper, and a *clean* side-arm vacuum flask attached to a vacuum line. It is important to add a little water to the filter paper with the vacuum applied so the filter paper sticks to the funnel.

2. As the ***filtrate*** (the stuff on top) begins to dry out, rinse three times with 1-3 mL of water, continuing to pull the vacuum.

3. Transfer the filtered solution to a 50 mL volumetric flask using a funnel. After pouring the solution into the flask, rinse three times around the side-arm flask with about 1-3 mL of DI water, emptying the rinse water into the volumetric flask each time. Then rinse the funnel. This is called a ***quantitative transfer***, as it attempts to get all of the material into the new container. This should be your normal practice for transferring liquids between containers.

4. Bring the volume of the solution up to exactly 50.0 mL using the pH 5 acetate buffer. This solution is your ***reagent solution.***

***Standard solutions (per table):***

5. *Carefully* measure 12.5 mL of 100 ppm zinc sulphate with a graduated cylinder.

6. ***Quantitatively transfer*** to a 50 mL volumetric flask and bring the total volume to 50.0 mL using the pH 5 acetate buffer. The buffer is This new solution is 25 ppm.

7. Repeat steps 5 and 6 with 2.5 mL, 5.0 mL, 7.5 mL, and 10 mL. What are these solution concentrations (hint: how does 2.5 compare to 12.5)?

***Standard curve (groups of 2)***

8. Using a calibrated micropipette, combine 1 mL of the 25 ppm solution with 5 mL of reagent solution. Use different tips for the zinc sulphate and the reagents solution. Do not touch the mixture while adding the reagents solution. *After* adding the last mL of reagent solution, pipette up and down several times to mix.

9. Fill a cuvette 2/3 from the top with this mixture. Fill 1 mL at a time, and make sure you know exactly how many mL you used for the cuvette.

10. Connect a SpectroVis to a LabQuest. Make sure the units are set to "Abs". If they aren't, click on the red banner and select "Change Units".

11. Click on the red banner and select "Calibrate". Follow the instructions on screen.

12. Click on the red banner again and select "Mode". Choose "Full Spectrum".

13. Use the "play" button to start taking data.

14. The best place to get data is usually the wavelength where absorption is the highest, the $\lambda_{max}$. What is the $\lambda_{max}$?

15. Return to the home screen and click on the red banner again and select "Mode". Choose "Events with Entry", and set the wavelength to $\lambda_{max}$. For the event, give the label "[Zn]" and the unit "ppm".

16. Use the "play" button to start taking data.

17. Place your lowest concentration sample in the reader, making sure the clear sides line up with the light and arrow icons on the SpectroVis. Hit "Keep" on the bottom of the screen, and enter the concentration for the first event. Repeat for each sample, moving from lowest concentration to highest. Keep your highest concentration (25 ppm) cuvette.

18. Hit "stop" to complete data collection.

19. Use the "Table" icon to navigate to your data in table form. Download a ***local*** copy of the linked Excel spreadsheet and record the absorbance data on your spreadsheet.

20. Create a scatterplot with Target concentrations (A5-A9) as the x-axis and Absorbance (E5-E9) as the y-axis. Make sure to label axis and the graph clearly.

21. Add a linear-trendline for the displayed data, showing the equation and $R^2$ value. What is the molar absorptivity ( ) of the acacia / zinc solution at 550 nm?

***Evaluation of remediation candidates (groups of 2)***

22. Ask your instructor for a remediation candidate. You will be given a concentrated solution of glutathione, cysteine, selenium yeast, garlic extract (use a hood), brewer's yeast (which all have sulfur-containing compounds), and citric acid, alginic acid, succinic acid, and nitrilotriacetatic acid, which are all carboxylic acids.

23. Return to the home screen and click on the red banner again and select "Mode". Choose "Events with Entry", and set the wavelength to $\lambda_{max}$. For the event, give the label "Remediation Compound" (use the name of your remediation compound) and the unit "drops".

24. Use the "play" button to start taking data. Hit keep immediately and enter "0" for the number of drops.

25. Add a drop of your remediation compound. Use a separate pipette to mix the solution by pipetting up and down. Check to see if your absorbance has changed significantly. If it has, hit “Keep” on the bottom of the screen, and enter the number of drops for the first event. If it hasn’t, slowly add (while counting) drops until you’ve noted a change, then hit “Keep” and enter the number of drops. Continue adding drops in the same increment you chose between the first and second number of drops 5 more times, keeping and recording the number of drops each time.

26. Hit “stop” to complete data collection.

27. Use the “Table” icon to navigate to your data in table form, and record the absorbance data on your spreadsheet by changing the sheet to “Evaluation of Remediant”.

28. Using your calibration curve’s line of best fit equation (step 21), create a calculate column in Excel for the concentration of free zinc corresponding to each absorbance measurement (C5-C11).

29. Create a scatterplot with “drops of remediation candidate” (A5-A11) as the x-axis and concentration of zinc (C5-C11) as the y-axis. Make sure to label axis and the graph clearly.

30. You do not need to create a line of best fit for this data.

31. On the ***online*** version of the linked Excel spreadsheet, go to the “Class Data” sheet and enter your names, remediant, and data in the next available slot. It should automatically plot in the “Plot of Class Data” tab.

**Post-lab questions:**

16. Look at the “Plot of Class Data”. What types of remediants were most successful? Least successful? How can you tell?

17. Propose an experiment that you would like to perform to follow up now that you’ve seen some preliminary data.

**See also:**


· W. Siriangkhawut, K. Ponhong, and K. Grudpan, A green colorimetric method using guava leaves extract for quality control of iron content in pharmaceutical formulations, *Malaysian Journal of Analytical Sciences*, ***23***, 595–603, (2019).

· El-Khishin, I. A.; El-fakharany, Y. M. M.; Omaima I. Abdel Hamid, O. I. A.; Role of garlic extract and silymarin compared to dimercaptosuccinic acid (DMSA) in treatment of lead induced nephropathy in adult male albino rats, *Toxicology Reports*, ***2***, 824-832, (2015).


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# 5: COMPOSITION OF MOLECULES AND SOLUTIONS

## LAB 5: PERCENT COMPOSITION AND MOLECULAR FORMULA

Bonjour, J. L.; Pitzer, J. M.; Frost, J. A. *J.Chem. Ed.* **2015**, *92* (3), 529-533.

**Introduction:**

In class, we are beginning to look at the different ways atoms can bond together to form molecules and the relationships between formulas and covalent structures. One of the most significant ways covalent structures are determined for organic molecules is by the means of Nuclear Magnetic Resonance (NMR) Spectroscopy. $^1$H (Proton) NMR is often used to determine where hydrogen atoms are relative to each other and to other atoms. We will use Proton NMR in this lab to introduce you to the principles of the instrument and explore the relationships between structure, formula, and chemical signal.

Proton NMR gives information about each distinct proton (identical protons give identical signal) in three different ways

1. Shift (where the signal shows up based on what is near to the proton)

2. Integration (how many protons are giving the signal relative to other signals)

3. Splitting (how many different protons are nearby)

For the purposes of this lab, we only need to identify factors influencing shift and integration. We will use these two factors to separate signals we get from components in a solution. In this case we will be using rubbing alcohol, which is a mixture of isopropyl alcohol as the major component (solvent) and water as the minor component (solute). As these signals are based on the ratio of atoms, they can be used to give molar ratios of components. However, most common solutions are reported as percentages by mass or by volume (m/m% or v/v%), molarity (moles/liter), parts per million (ppm) or parts per billion (ppb).

Mouthwash, vinegar, and rubbing alcohol all report percentages by volume. We can use the technique in this lab to determine the percentage by volume for solutions that aren't reported as well. For example, we have used this technique to verify ethanol and water ratios in imitation almond extract, and determine the relative concentrations of the benzaldehyde that makes it almond-y.

In this lab you will determine the concentration (as a percentage by mass and volume) of a sample of rubbing alcohol. You can find a variety of concentrations available in most pharmacies.

Terms to remember:

· **Solution:** A mixture in which a solid, liquid, or a gas dissolves in a liquid. Aqueous (aq) solutions are water-based solutions.

· **Solvent:** The majority component of a solution. *In aq. solutions with concentrations < 50%, the solvent is water.*

· **Solute:** The minority component of a solution. *In aq. solutions with concentrations > 50%, the solute is water.*

· **Mass percent (m/m%)** = mass solute / (mass solute + mass solvent) x 100

· **Volume percent (v/v%)** = volume solute / (volume solute + volume solvent) x 100

**Procedure**

1. Obtain an unknown sample from your instructor.

2. Follow the directions for acquiring a NMR spectrum of your sample.

3. In the processing of your spectrum, note the following:

a. Do all the peaks you predicted show up? How do you know which peak is which?

b. The split peaks will all correspond to the organic component of your solution. Are any of these peaks overlapping with other peaks? Which one is the most clearly separated? Integrate as your reference peak.

c. Integrate all remaining peaks as one region.

4. Take a picture of your spectrum to use for the analysis portion of the lab and insert it your spreadsheet.

**Analysis –** This lab workup is a continuation of the pre-lab questions and uses the same spreadsheet.

1. Indicate the location of the peak you chose as a reference peak in ppm units (B36).

2. Indicate the integration of the peak you chose as a reference peak (B37).

3. Indicate the # of hydrogens represented by this peak based on the structure (B38).

4. The integration of the reference peak divided by the number of hydrogen atoms it represents = the number of this type of molecule represented in this spectrum. Calculate the number of (organic) molecules represented by the spectrum using a formula in B39.

5. The number of organic hydrogens not represented is the total number of hydrogens in the formula for the organic molecule of interest minus those represented (B38). Enter this number into B40.

6. The organic hydrogens that aren't in the reference peak show up as part of the other integrated area. This contribution is the number of organic hydrogens not represented (B40) times the number of organic molecules represented in the integration (B39). Calculate this using a formula in B41.

7. Enter the total integration of the other peaks in B43.

8. The contribution from water is the total integration of the other peaks minus the contribution from the main organic molecule. Calculate this in B44.

9. Enter the number of hydrogen atoms in water in B45.

10. The integration contribution from water divided by the number of hydrogens in water is the number of water molecules represented by this spectrum. Calculate this number using a formula in B46.

11. Calculate the ratio of organic molecules to water molecules using a formula in B47.

*Work out the following conversions on a separate sheet of paper. You may do all the calculations on paper (show your work), or just work out the unit conversions on paper and do the math in Excel. Make sure to clearly label your work in either case.*

12. The units for this ratio are $\frac{\text{molecules of organic component}}{\text{molecules of water}}$. Work out the unit conversions using dimensional analysis to $\frac{\text{moles of organic component}}{\text{moles of water}}$. Hint: Avogadro's number.

13. Now work out the conversion from $\frac{\text{moles of organic component}}{\text{moles of water}}$ to $\frac{\text{grams of organic component}}{\text{moles of water}}$, then to $\frac{\text{grams of organic component}}{\text{grams of water}}$. Hint: What property from your pre-lab has units containing grams and moles?

14. You can envision this result as $\frac{\text{x grams of organic component}}{\text{1 grams of water}}$, where the number you calculated in 13 is "x". Look at the definition of mass percent. How can you find the mass percent from using this number?

15. Next work out the conversion from $\frac{\text{grams of organic component}}{\text{grams of water}}$ (from 13) to $\frac{\text{mL of organic component}}{\text{mL of water}}$.

16. You can envision this result as $\frac{\text{x mL of organic component}}{\text{1 mL of water}}$, where the number you calculated in 15 is "x". Look at the definition of volume percent. How can you find the volume percent from using this number? This number will have some error because sometimes there is a change of volume upon mixing.

17. Compare your calculated value for volume percent to the reported value. What is the percent error?

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# 6: AVOGADRO COMPUTER MODELING

## LAB 06 – AVOGADRO COMPUTER MODELING

**INTRODUCTION**

Lewis dot structures are effective at showing connectivity between atoms, but to visualize arrangements in 3D, it is helpful to use molecular modeling software. Avogadro is a free tool (https://avogadro.cc/) to build and manipulate molecular models. It is available for most operating systems. The native functionality of Avogadro can fill out octets of molecules automatically, and optimize molecules on the basis of molecular mechanics (VSEPR and valence bond theories). It can also be used, as we will see next week, as a way of generating input for computational chemistry packages that allow access for better optimizations and better data.

**PROCEDURE**

**Part A: Using Avogadro**

1. Read through the instructions for drawing a molecule on the Avogadro website: https://avogadro.cc/docs/getting-started/drawing-molecules/. If you run stuck anywhere, it is good to return to this resource.

2. Open Avogadro and familiarize yourself with the tool bars just under the pull down menus. Besides the typical New, Open, Save, Close, & Quit tools are more important ones that work with your molecules. Hover the mouse over them to see what the Draw tool, Navigation tool, Bond Centric Manipulation tool, Manipulation tool, Selection tool, Auto Rotation tool, Auto Optimization tool, and Measurement tool each does. Click on the Tool Settings and Display Settings buttons to see how the application window changes. When both are unselected, the drawing window is bigger. The tool settings change as you click on the tools, and the display settings help to easily change how molecules are displayed.

3. Make sure that the draw tool is selected. Move your cursor onto the page and click the left mouse button to place the element. The type of atom that is selected in the draw settings will be placed at the cursor. If the Adjust hydrogens box is checked then some hydrogens have also been automatically added as well. This is usually helpful, but not always, so be careful. If you have not already gone through the first three step-by-step walkthroughs on the Avogadro website (Getting Started, Drawing Molecules, & Making Selections), do so now as these cover important skills that you will use frequently in this program. In the third tutorial, ignore the parts about SMILES, SMARTS, Residues, and Project Trees.

4. Now that you know how to build molecules and optimize their geometry, let's build some. Build a water molecule. Click on the navigation tool and rotate the water molecule to see how this tool works. In the absence of a middle mouse button use shift-up/down arrows for zooming.

5. Skip the Bond Centric Manipulation tool and instead select the Manipulation tool. Click and hold on an atom and move the mouse to see how this affects your molecule. Click and drag in the window but not on an atom to see what happens. You have already played with the select tool in the tutorials. We will skip the auto rotation tool.

6. We have a water molecule but it isn't very pretty. Its bond lengths and bond angle are incorrect. But that is what Avogadro is for. But before optimizing our molecule, let's characterize what we have so far. Click on the Measurement tool icon and then select a hydrogen atom, then the oxygen, and finally the second hydrogen. The order is important to match the values that appear on the screen. The first distance is between atoms #1 & #2 and the bond angle has the second atom at its apex. Selecting the atoms in another order would make these measurements meaningless. Download a local copy of the linked Excel spreadsheet and record these values for water under Pre-3D Optimization. Then reset the atom selection by clicking in an empty part of the window and select your three atoms in an odd order to see the distance and angle values that are displayed, and record these as well. Can you see how one needs to be careful? Finally, reset the atom selection and click only on the two hydrogen atoms to measure and record the distance between them.

7. Now to obtain an accurate 3D view; under the Extensions drop down menu select **Molecular Mechanics / Setup Force Field**. Select MMFF94 as the Force Field and click OK. Again, under Extensions, select **Optimize Geometry**. Your molecule's geometry is now optimized and arranged as it would appear in 3-dimensional space. Measure all interatomic distances and its bond angle as above and record in the optimized column of your Excel sheet.

8. With the water molecule in the center of the window, select display settings and explore the Display Types but be prepared to restart the program as several of these options crash it. Notice that the ball & stick rendering (The different ways that a molecule can be displayed are known as renderings.) is the default. Uncheck this type and select each of simple wireframe, stick, Van der Waals spheres, & wireframe renderings. Notice that some display types have wrench icons. That means that some details of their displays can be changed by the user.

9. Use ctrl-backspace to remove your water molecule. Now build each of the following molecules: carbon dioxide (O=C=O), nitrogen (NºN), and ammonia ($NH_3$). Repeat step five for each one to get pre- and post- geometry optimization characteristics as you did for water. Record your measurements.

10. It's time to compare our theoretical calculations and geometries with experimentally derived ones. A good web site that contains experimental parameters of many molecules is http://cccbdb.nist.gov/. We will come back to this web site later, so it would be a good idea to bookmark it or put it in your favorites for easy access. Click on the "Experimental" tab on the top, select all experimental data for one molecule, and enter water (or $H_2O$) as the molecular formula. Scroll down to the geometry data and record the actual experimental values

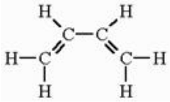
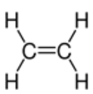
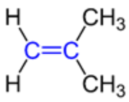
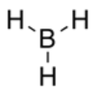
next to the information you got from Avogadro. Record these in your document. Repeat for the values that you recorded for carbon dioxide, nitrogen, and ammonia.

11. Answer the 3 questions listed below the data tables in the linked spreadsheet directly in Excel.

Remember, we have used a quick geometry optimizer in Avogadro for displaying molecular geometry. It uses a process known as MMFF94 that is a **molecular mechanics force field** method that has been optimized for organic compounds, specifically drug compounds. It is based upon estimating forces between atoms in a molecule and so is called a **force field** method. These work well for quick geometry optimizations, but when more accurate geometries are needed for more accurate calculations, they do not produce acceptable results, and we must resort to more complicated methods. Avogadro serves as a front end for these calculations. There are many computational packages that can perform these well, but the free ones are not user friendly. You will submit a few optimized Avogadro molecules and your instructor will run them so that you can see the usefulness of these computations.

Along with the Excel sheet, please submit Avogadro save files for the following molecules. Your instructor will return molecular orbital diagram results for you to use in next week's lab.

Butadiene Ethene Isoprene Borane

# 7: MOLECULAR ORBITALS

## LAB 07 – MOLECULAR ORBITALS

**INTRODUCTION**

In the previous lab, we used molecular modelling software (Avogadro for most of you) to quickly build structures of molecules that would show reasonably correct molecular geometries, including shapes, bond lengths, and distances. We compared the values generated to experimental data, and probably found that molecules were fairly accurate when simple, but errors increased with complexities like more atoms and pi (double) bonds.

The molecular models used in Avogadro and other free modeling tools are generated and optimized using molecular mechanics – in the very simplest version each atom is treated as a single point with a defined charge, and each bond is treated like a spring. Atoms are not single points, they don’t always have fixed charge, and their bonds can be influenced by the rest of the atom. All of these assumptions lead to cumulative errors that can be fairly significant in certain cases.

A much more robust way to model data is to use Molecular Orbitals (MOs), the simplest versions of which we talked about in Chapter 5.4. Molecular Orbitals, like the atomic orbitals we talked about in Chapter 3.1-3.3, must consider how every single electron in a system affects every other electron. We saw that this was impossibly difficult to accomplish with anything more certain than probability densities as soon as we had two electrons, like in Helium. Molecular orbitals generally have at least two electrons and two atoms and are even more complex. We use computational packages to generate molecular orbital information by solving Schrödinger’s equation under a set of assumptions. The nature of these assumptions and the mechanics of the calculation are the subject of Physical Chemistry II or Modern Physics, *but* we can run the packages and use the orbitals generated to inform our understanding of chemistry right now.

Obviously, we can’t run complex computations every time we want to see whether an atom is likely to have a dipole. Even though we can get better information from Molecular Orbital Theory, models like Valence Bond Theory give us answers that are relatively close most of the time, and *way more accessible*. Molecular Orbital Theory gives us much better answers when we want to talk about bonds broken and bonds formed, charged species, and anything that has pi-bonds, but we can only use it qualitatively without calculations, like we did in chapter 5.4. This lab gives a first glimpse at how useful the information given by Molecular Orbital Theory can be.

Molecular orbitals are often used to predict reaction energies and directions. Reactions occur when electrons move from one orbital into another to form a new bond. The electrons that are most reactive are the highest energy electrons. We talk about the orbitals that contain electrons as *occupied* molecular orbitals, the highest energy electrons then are in the **Highest energy Occupied Molecular Orbital (HOMO).** The orbitals that are empty (*unoccupied* molecular orbitals) are able to receive electrons, with the **Lowest energy Unoccupied Molecular Orbital (LUMO)** being the most reactive. This leads to a few key points:

- In general, reactivity can be explained by electrons moving from a HOMO to a LUMO that it can overlap in symmetry. You can see in the figure below that a reaction is allowed if the overlap would be all in phase between overlapping orbitals, but if some would be in phase and some would be out of phase, it would not be allowed.
- When a bonding orbital is the HOMO and contributes electrons to a LUMO, a new bond is formed and the bonding orbital that was the source of the electrons is now empty (a bond is broken).
- When the LUMO is an antibonding orbital, the bond(s) that it is associated with are broken when electrons are added from the HOMO. In the example of the next page, the C-Cl bond is responsible for the LUMO, and will be broken when $HO^-$ reacts.

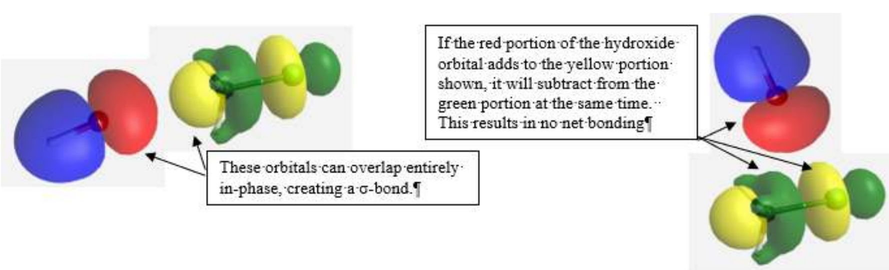



**FIGURE GENERATED USING WEBMO**

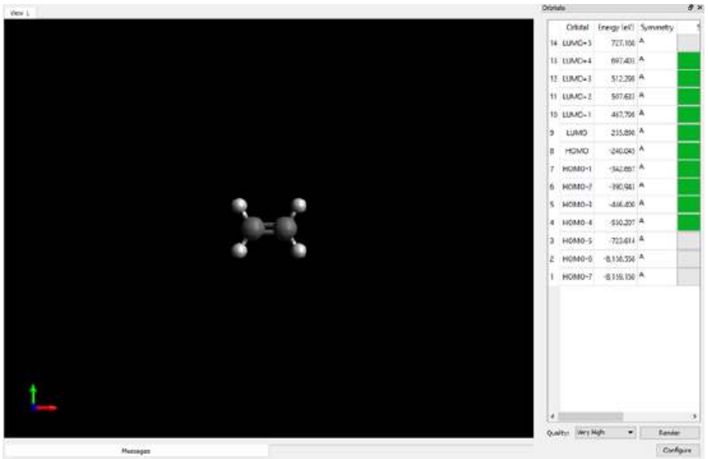
Polik, WF, Schmidt, JR. WebMO: Web-based computational chemistry calculations in education and research. *WIREs Comput Mol Sci*. 2021;e1554. https://doi.org/10.1002/wcms.1554

The interactions between atoms are defined by symmetry – interactions like those shown between the $HO^-$ and the $CH_3Cl$ on the above left are all one phase along the bond axis, and are considered to have sigma symmetry (think “s” orbital). Sigma symmetry can interact with symmetry. If the $HO^-$ is trying to interact with the LUMO from on top, as shown in (b), the C-Cl bond has two different phases along the proposed interaction (pi symmetry, like a p-orbital).

In Avogadro, all occupied orbitals get displayed as “HOMO -X”, with the actual HOMO labeled “HOMO” and the next highest energy labeled “HOMO-1”, then “HOMO-2”, etc.

Similarly, all unoccupied orbitals get displayed as “LUMO +X”, with the actual LUMO labeled “LUMO”, and the next lowest energy labeled “LUMO+1”, etc.

In this lab, we will be trying to identify the symmetries of HOMO and LUMO interactions in order to understand the geometry of some common reactions in organic chemistry.




Start by opening the file for ethene, $H_2C{=}CH_2$. These files are all generated as “.log”, which will probably not default to opening in Avogadro. It is easier to open Avogadro first and then choose the file. When the file opens, you should have a screen similar to the screen on the right. The HOMO and LUMO and several orbitals above and below are shown in green, which means that the file has already rendered (drawn) those MOs. If you click on that row, the MO will display over the ball and stick model. You can show other rows by clicking them and then clicking “render” on the bottom right. Play around with the controls a little bit and then watch the video attached to this lab about interpreting the MO diagrams.

Fill out the information requested for the *actual* HOMO and LUMO in each molecule. The first row follows the discussion in the video.

| ***Molecule*** | ***HOMO Energy*** | ***LUMO Energy*** | ***HOMO Type*** | ***LUMO Type*** |
|---|---|---|---|---|
| *Ethene* | -240.045 | 235.898 | $\pi$ | $\pi*$ |
| *Butadiene* | | | | |
| *Isoprene* | | | | |
| *Borane* | | | σ* | p |

1. Ethene and butadiene react via a reaction process called a “Diels-Alder Reaction”. Compare the energy differences of the LUMO of ethene and the HOMO of butadiene vs the LUMO of Butadiene and the HOMO of ethene. Which is more significant? Which interaction is more likely (Hint: less energy is better and faster)?

2. If adding an atom or group of atoms to butadiene decreased the energy of its LUMO, do you think this reaction would be faster or slower? Why or why not?

3. Draw the LUMO of butadiene or printscreen with a view that clearly shows the symmetry (probably from the side). Draw the HOMO of ethene or printscreen with a view that shows the symmetry (again probably from the side).

4. The first and last carbons for butadiene will interact with the carbons of ethene from either above or below (not straight alongside, since the hydrogens are in the way). Based on the orbital pictures from 2, what symmetry do you expect the interactions to have (remember – sigma has no nodes in the direction of interaction for a circular cross-section, pi has one node for a hamburger-shaped cross-section)?

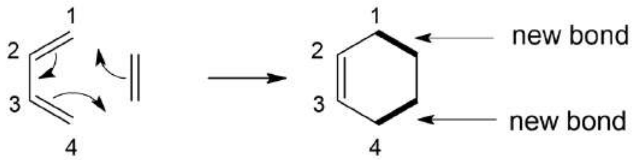
5. What type of bond would form between the end carbons of the molecules (same as last answer)?

6. The LUMO of butadiene has both bonding and antibonding intereactions. Labelling the carbons 1-4, which carbons have antibonding (out of phase) interactions? Which carbons have bonding interactions?

7. Which bonds do you expect to be broken? Where should there be a new bond formed?

8. The pi electrons in the ethene are part of the HOMO. What does this predict about the second bond of ethene (see bullet points at end of first page)?

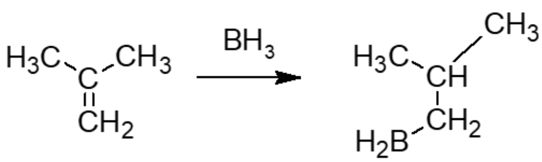
9. The Diels-Alder reaction and product is shown below. Does the product shown match what you’d expect for your answers to questions 4-8?




10. For the next questions we'll be looking at the molecules $BH_3$ (borane) and isoprene. Compare the energy differences of the LUMO of borane and the HOMO of isoprene vs the LUMO of isoprene and the HOMO of borane. Which is more significant? Which interaction is more likely (Hint: less energy is better and faster)?

11. The LUMO of borane is more or less an empty p-orbital. Isoprene's HOMO is a full pi-orbital, but its shape is influenced by the adjacent $CH_3$ groups. Based on the shape of the HOMO, which carbon of isoprene's double bond has the most electron density with which to interact with borane?

12. The reaction of a double bond, such as that in isoprene is called a "hydroboration" and is fairly selective for which carbon of the double bond gets the boron. A reaction scheme is shown below. Does it match your expectations based on 10 and 11? Why or why not?




This lab is intended to get you thinking about how electron spaces like molecular orbital can shape the reactivity of molecules in preparation for organic chemistry next year. Anytime we use MOs for now, we will continue to walk through each step – what are the HOMO and LUMO that would likely interact? With what symmetry would they interact? Which areas of the molecule are most reactive? How do we see these in the product? We won't use these except in guided settings until organic chemistry.

# 8: VERY GREEN FLUORESCENCE

## LAB 08 – VERY GREEN FLUORESCENCE

Inspired by *Applied Surface Science Advances 13 (2023) 100385*

**About Fluorescence**

In labs 2 and 3, we identified absorbance of visible light as a useful way to measure concentrations in solutions using standard curves for both molecular absorption and atomic emission. Standard curves can be generated for any method that predictably relates a ***measurement***, like absorbance at a specific wavelength, to ***determine*** a property, like concentration. Many of these happen without us even realizing it – scales relate the amount of force needed from an electromagnet to offset the pull of gravity, which is measured by the amount of current drawn for the electromagnet. So a scale actually ***measures*** the current needed to match the weight of gravity, but it is calibrated so we can make a ***determination*** of the mass.

Often the types of signals we can get in a solution are complex. When we generated a standard curve for fluorescein absorbance in Lab 2, we only had fluorescein, sodium hydroxide, and water in solution. When we generated a standard curve in Lab 4, we had a more complex indicator, but a clear zinc control. But the world is full of colored compounds – how can we make sure that signal only comes from a molecule we are interested in (***analyte***) instead of one of those compounds?

We generally look for relationships that are ***selective*** – giving more signal for one analyte than others – or, even better, ***specific*** – giving signal for only the analyte of interest. For example, the thin lines of atomic emissions are often fairly specific to an element if the instrument has good resolution between wavelengths. On the other hand molecular absorptions of the chelators like the catechols we used in Lab 4 are formed only when there are metals in solution but give signal for many metals making them selective towards metals but not specific.

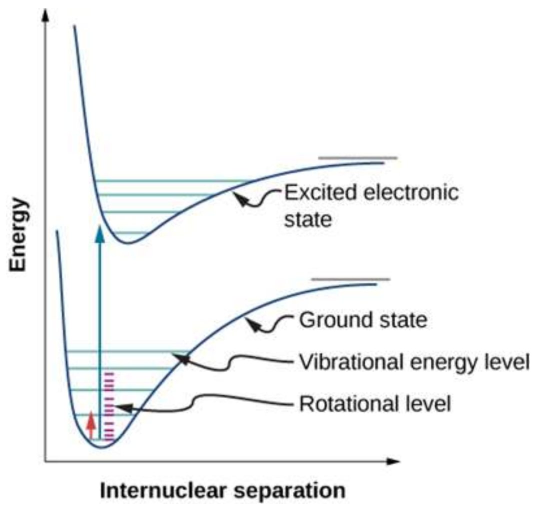



Figure 1: Energy changes during molecular spectroscopy are dependent on electronic transitions, and to a smaller degree vibrations and rotations. (CC BY 3.0; OpenStax)

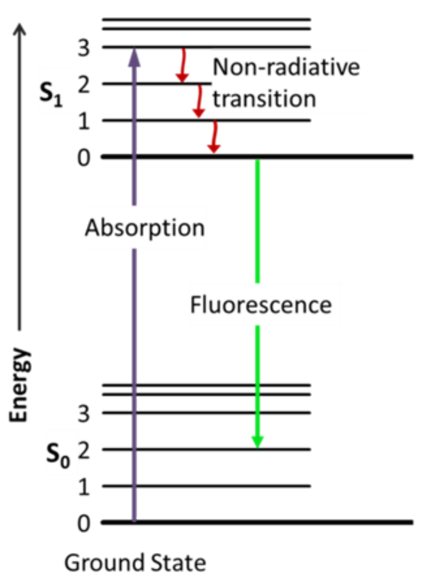


Figure 2: Jablonski diagram of absorbance, non-radiative decay, and fluorescence. (Public Domain, Jacobkhed via LibreTexts.org)

Absorbance of ultra-violet or visible (UV-Vis) light is an incredibly common phenomenon, with many compounds having absorbance at any given wavelength. This is because the UV-Visible region of light matches with the amount of energy required to move electrons between the HOMO and an unoccupied orbital of many organic molecules and inorganic compounds. The breadth of the signal in molecular UV-Vis absorptions is because there are slight changes in the energy required as a molecular bonds vibrates or rotates. Recall that the energy stored in a bond depends on the distance between atoms as shown in **Figure 1**. Atomic absorptions tend to look more like lines because atoms don't have bonds to vibrate. In **Figure 2**, the difference in energy between the singlet state (S, indicating no unpaired electrons) HOMO and unoccupied orbital is indicated by $S_0$ and $S_1$, while each of the lines 0, 1, 2, 3… indicate the vibrational states. Typically, when a molecule absorbs energy, it can absorb energy to increase both the electronic state energy (ie $S_0$, $S_1$, etc.) as well as the vibrational state. When many vibrational modes are available, relaxation often occurs quickly through vibrational states, and energy is released as heat. However, when there are few vibrational states available, like in rigid circular molecules, relaxation occurs rapidly within available states, followed by relaxation through the emission of a visible photon to reach an electronic ground state ($S_0$), a phenomenon known as fluorescence. Since some energy is lost to vibration, the energy of the fluorescence is typically less than that of absorption and the wavelength is longer.

Luminescence is a general term used to describe the emission of light (a photon) of some wavelength as a material relaxes from an electronic excited state back to the ground state. There are many ways a material can be excited, but in the context of this experiment we are particularly interested in photoluminescence (or PL), where excitation is the result of absorption of a photon of some, usually shorter, wavelength.

Now, why does any of this matter? The usefulness of fluorescence spectroscopy stems from two main factors. First, since most molecules do not exhibit fluorescence the background fluorescence signal is often very small. Therefore, the technique can detect specific molecules with high sensitivity. Second, the spectral properties of the fluorescence emission are usually highly sensitive to local environment (solvent, pH, ionic strength, etc.). Thus, the use of fluorescence is a key technology in applications where local environments can vary quite dramatically, for example in the study of macromolecule systems and biological specimens.

Chlorophyll, well known for its light absorption properties, has fluorescent properties. There are several types of chlorophyll. We will be measuring chlorophyll A (**Figure 3**), from an undifferentiated standard because it absorbs light in the right range for our fluorimeters.

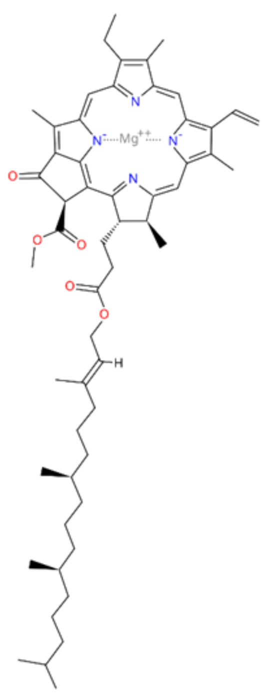


Figure 3: Chlorophyll A has an extended network of π bonds that create a fluorescing rigid structure.

**Procedure:**

1. Choose a green you want to test. Use a scissors to cut 10 grams of material it into small pieces. Grind in a mortar thoroughly.

2. Add 30 mL of 95% ethanol to the mortar and briefly grind.

3. Vacuum filter using a Buchner funnel into a clean side-arm flask.

4. After the material has passed through the filter, remove from the vacuum immediately and transfer into a 50 mL beaker. Cover with parafilm when not in use.

5. Prepare a chlorophyll solution from 0.500 g of chlorophyll dissolved to 1.000 L in 95% ethanol. Calculate the concentration of this buffer in µg/cc (the units from the original literature) and record this as the *stock standard* concentration in your local copy of the linked Excel

spreadsheet.

6. Prepare a 100-fold dilution of the stock standard into a 95% ethanol. You will probably want to pour out a small amount of the *stock standard* into a beaker to make measuring easier. You want your final volume to be 100 mL in a volumetric flask. Calculate the concentration of this buffer in units of µg/cc. Show your work in your Excel spreadsheet. This is your *working standard.*

7. Setup Vernier GoDirect SpectroVis Plus by plugging the USB into the instrument and your LabQuest. On the LabQuest, click on the red banner and select “Change Mode”. Select “Fluorescence”. For excitation wavelength, select 405 nm. For emission wavelength, select 674 nm. The SpectroVis may alter this slightly to a number near 674 nm. This is normal and not a concern, but you should make sure to take note of the actual number.

8. Obtain a plastic fluorescence cuvette, touching only the very top, and fill about ¾ full with 95% ethanol, and carefully place in the SpectroVis so that the clear sides are lined up with the light and detector. Calibrate the spec by selecting the following menus/buttons with the stylus (NEVER A PEN OR PENCIL) (Sensors / Calibrate / USB:Spectrometer / {wait for the warm up period} / Finish Calibration / OK). Before your begin to make measurements, go to the gear on the top righthand side of the window. Place your blank (either buffer or water) in to the cuvette holder, select an integration time of 100 ms (this increases the signal intensity by adding the signals over this time), and temporal averaging of 5 (this reduces the variability by taking replicate measurements automatically), and confirm excitation wavelength as 405 nm.

9. Label a white piece of paper with numbers 1-6, spaced enough that each number could have a cuvette placed on top. Using a calibrated micropipette, add the following amounts of ethanol and your *working standard* chlorophyll solution to create cuvettes 1-6.

| **Cuvette** | **1** | **2** | **3** | **4** | **5** | **6** |
|---|---|---|---|---|---|---|
| Water (mL) | 0.0 | 1.0 | 1.5 | 1.8 | 1.9 | 1.95 |
| Chlorophyll (mL) | 2.0 | 1.0 | 0.5 | 0.2 | 0.1 | 0.05 |

10. Remove the blank cuvette. Test the fluorescence intensity of each cuvette. Record your data into your local copy of the linked Excel spreadsheet. You will have to calculate out the concentration of each cell.

11. Create a linear regression plot for the standard curve. Add a trendline and show both the equation and R-squared value.

12. Take a fluorescence measurement of your extracted plant chlorophyll. Does the fluorescence intensity fall into the range of your standards? If so, proceed to step 15. If not, dilute 10-fold and repeat this step. Continue diluting and measuring your sample until it falls within the range of your standards. Make sure you record how many dilutions you needed! Record this data under “Sample Fluorescence”

13. Now that you know how many dilutions you need, take two more samples and dilute them to the proper concentration for measurement, for a total of 3 measurements of 3 distinct samples of your chlorophyll (not the same sample 3 times). Convert each to concentrations using your standard curve, then calculate the average and standard deviation for your samples only.

14. If you diluted your sample, find the average concentration of the original, undiluted sample. Note that the standard deviation should scale by the same value. Record both in the section “Values for undiluted samples:”

15. Record your average and standard deviation on the ***online*** copy of the linked Excel spreadsheet on the “Class Data” sheet.

**Clean up:**

All solutions should be dumped into the large “Green Clean” collection beaker in the hood.

**Post-lab questions:**

1. Using the data shared by your classmates for the same plant, calculate the average value and the standard deviation. Is this technique precise?

2. Your concentrations are in µg (of chlorophyll) / cc (solution), but most analyses of extracted analyte want to know how much you had in the original mass of sample. Could you calculate this directly from your data? There is a major source of error if we tried to analyze this way. What is it?

3. Assume that you were able to recover all of the 30 mL of 95% ethanol you originally added. What is the concentration of chlorophyll per gram of your plant?

4. Can you think of a way to make sure you know the volume of solvent that has all your extracted chlorophyll? What would that look like in the experiment

## 9: Iron Determination by Titration

**INTRODUCTION**

Quantitative analysis is the measuring of the amount of a substance in a mixture. There are multiple standard quantitative techniques which can be used to measure concentration, and one that we are going to learn about today is titration.

Titration works like this:

a) find a compound (the titrant) that reacts with the compound (the analyte) that you want to measure in a mixture,

b) determine the stoichiometry of this reaction,

c) make up a solution of the titrant whose concentration is accurately known,

d) add just enough titrant to balance the amount of analyte in your unknown sample,

e) from the volume of titrant, concentration of titrant, and reaction stoichiometry, calculate the amount of analyte originally present in the sample.

Two problems have to be overcome for a titration to work. First, the concentration of the titrant needs to be accurately known. Often this solution is not easily made to an accurate concentration. Thus, it needs to be standardized (its concentration needs to be accurately determined). Secondly, how do we know when just enough titrant has been added to react with the amount of analyte (the equivalence point) in the unknown sample? Often we can add an indicator which changes color as soon as there is an excess of titrant in the solution which is defined as the end point. The end point is usually just after (but quite close to) the equivalence point. Thus, we will take the end point to represent the equivalence point.

In this lab, we will measure the amount of analyte, that is iron II ($Fe^{2+}$), in a solution by measuring the amount of an oxidizing agent, $KMnO_4$, that it takes to oxidize it to iron III. Potassium permanganate easily oxidizes iron II to iron III by taking an electron from each iron II cation. Obviously, the stoichiometry is not one-to-one. The following is the balanced reaction for the reduction of the permanganate anion under acidic conditions:

8 $H^+$(aq) + $MnO_4^-$(aq) + 5 $e^-$ à $Mn^{2+}$(aq) + 4 $H_2O$

So, one must determine how many iron II atoms need to be oxidized for the reduction of each permanganate anion. You will determine this ratio in your pre-lab assignment.

Now how do we know when we have added an equivalent amount of titrant to just match the iron II in our sample? This is quite easy because the permanganate ion ($MnO_4^-$) is dark purple, but the reduced manganous ion ($Mn^{2+}$) is colorless. Thus, as you begin to add the titrant, the purple immediately disappears, but as soon as all the iron II is oxidized and a small excess of permanganate has been added, a faint purple persists in the solution. You want to stop adding the titrant just as the faintest purple (looks more like pink) persists in the solution. The amount of titrant added will be very accurately determined using a buret. The amount of titrant in the buret is recorded both before and after the titration. Therefore, the difference is the amount of titrant added to the reaction. Because this reaction produces a color change at the end point, we will not need to add an indicator in this experiment.

Permanganate solutions cannot be made up with an accurately known concentration; thus, each of us will help standardize the titrant at the beginning of the lab. Then each of you will titrate your unknown sample three times. You will use your results to calculate the amount of iron II present in your unknown.

**PRE-LAB QUESTIONS**

The following is the balanced reaction for the reduction of the permanganate anion under acidic conditions:

8 $H^+$(aq) + $MnO_4^-$(aq) + 5 $e^-$ à $Mn^{2+}$(aq) + 4 $H_2O$

What is the balanced reaction for the oxidation of iron II by permanganate under acidic conditions? Note this balanced equation will be needed in lab.

What is the ratio of $MnO_4$ to iron II? ________________________________

In today's lab, are we going to use the equivalence point or the endpoint? Explain why in your own words.

Why is it so important to look for the faintest color change that persists for 30 seconds? Explain in your own words.

**PROCEDURE**

Preparing for the titration

1. Obtain a buret and buret clamp and mount them on your stand keeping the buret vertical. Burets should be clean, but you should test them in any case. Close the stop cock, place a beaker under the tip, fill to close to the 0 mark with DI water, and open the stop cock to let the water empty into the beaker. If any water droplets remain on the inside of the buret, it is not clean. Titrant will be left behind during your titration, and the amount you determine for titrant added will be inaccurate. During lab we will discuss proper cleaning before and after use. Also, make sure that your buret does not leak.

2. Using 6 M $H_2SO_4$, prepare 300 mL of 1 M $H_2SO_4$. Download a *local copy* of the linked Excel Spreadsheet. In the Spreadsheet, calculate the amount of 6M sulfuric acid needed. First add the DI water that you determined was needed to make the 1 M $H_2SO_4$ to your 400 mL beaker, then add the amount of 6 M $H_2SO_4$ you determined was needed. (This will be accurate enough for this experiment.). Mix with your stirring rod.

3. Obtain an unknown iron II sample and record its label in your report sheet.

4. Obtain about 100 mL of titrant ($KMnO_4$), rinse your buret with this titrant as described at the beginning of lab, and fill your buret to close to the 0 mark, but do not go over. Make certain that all air/air bubbles from the tip of the buret have been eliminated.

Standardization of Potassium Permanganate

1. Using weigh paper, obtain about 0.6 g of ferrous ammonium sulfate hexahydrate ($Fe(NH_4)_2(SO_4)_2{\cdot}6H_2O$) and record the mass as accurately as possible. Record this and the following titration data in your local copy of the Excel Spreadsheet. This is the iron II standard with a molar mass of 392.2 g/mol. Add this solid to a 250 mL Erlenmeyer flask (washing all of it in with a little DI water from a wash bottle), add 40 mL of the 1 M sulfuric acid, and swirl to completely dissolve the solid. To prevent yellowing of the solution add about 3 mL of 85% $H_3PO_4$. This yellowing can mask the initial pink color at the end point.

2. Record the initial volume of titrant in the buret to hundredths of a mL. With swirling, add titrant to the sample until the faintest pink color possible persists for at least 30 seconds. Record the final volume of titrant in the buret.

3. Using the volume of titrant, moles of iron II standard, and stoichiometry, calculate the concentration of the titrant. Input this value into the ***cloud version*** of the linked Excel spreadsheet under sheet “Class data”.

4. After everyone has entered their values, copy the data into your *local copy* of the Excel spreadsheet and calculate the average and standard deviation for the concentration. You can begin your unknown titrations before you know this value.

Titration of Your Unknown Iron II Sample

1. Using weigh paper, obtain about 0.35 g of your unknown sample and record the mass as accurately as possible. Add this solid to a 250 mL Erlenmeyer flask (washing all of it in with a little DI water from a wash bottle). Add 40 mL of the 1 M sulfuric acid and swirl to completely dissolve the solid. Add about 3 mL of 85% $H_3PO_4$.

2. Record the initial volume of titrant in the buret, with swirling add titrant to the faintest pink endpoint and record the final titrant volume. See Excel sheet.

3. Repeat steps 1 & 2 with two more 0.5 g samples of your unknown.

4. Waste disposal: All purple colored solutions will go into the waste container labeled “permanganate waste”. All pink solutions can be poured down the drain.

5. For each of your three titrations, use the volume of titrant, standardized concentration of titrant, mass of unknown, and reaction stoichiometry to calculate the percent mass of iron II in your sample. See Excel sheet.

6. Calculate the mean and standard deviation for the percent mass of iron II from your three replicate values. Report these values on your Excel sheet.

**Final Questions**

1. How close was your standardized $KMnO_4$ concentration to the average of the whole class? Calculate the % difference between your value and that of the class (this is much like a % error calculation). %diff = (your molarity – class avg molarity)/class avg molarity * 100%

2. How would your results turn out if an additional unknown metal was present in the sample as a contaminant that is also oxidized by $KMnO_4$? Would your calculated result be lower, higher, or the same as the actual amount of $Fe^{2+}$ present? Why?

3. What are the standard deviations for the whole class standardization of the $KMnO_4$ and for your three titrations of your unknown? Show your math for calculating the standard deviation for your three replicates. What do these standard deviation values tell you about the precision or accuracy or both of the reported values for the $KMnO_4$ concentration and the % iron in your unknown sample?

4. This type of experiment can be quite accurate with proper care. List three steps that you think have the most potential for introducing error into your final result. Explain how each would add error to your results. Order them from most severe to least severe.

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## 10: Gas Laws

*Adapted from J. Chem. Educ., 2007, 84(3), pp 465-468*

INTRODUCTION

The pressure, volume, temperature, and amount of a gas are interrelated. Some of these relationships are directly proportional and some are indirectly proportional to each other. In this lab we will determine the relationship between the pressure and volume of a confined gas (temperature & moles remain constant) and then the relationship between the pressure and temperature (moles and volume remain constant). After taking measurements, you will record data, create graphs and determine the equation of the best fit lines on the graphs. Also from the data of the second experiment you will estimate absolute zero. If we assume that pressure is zero when all thermal activity ceases, then this is an estimate of absolute zero.

### PROCEDURE

**PART A. Setup & Data Collection for Pressure vs Volume (Moles and Temperature Constant)**

1. Connect Equipment
    - Turn on LabQuest.
    - Plug the pressure sensor into any channel (CH1–CH4).
2. Set Up LabQuest on the Meter Screen (has a red banner)
    - Tap the Mode button → Select Events with Entry.
    - Enter “Volume” for “Name”, and “mL” for “Units”, then tap OK.
3. Change Pressure Units to atmospheres (atm)
    - Go to Sensors → Change Units → (your channel) → Gas Pressure Sensor → atm.
4. Prepare Syringe
    - Set syringe to 5 mL (don’t push or pull after this).
    - Attach syringe gently by twisting onto the pressure sensor (snug, not tight!).

5. Start Collecting Data

- Tap the green Start button.
- One partner adjust the syringe to 2 mL, the other tap “Keep” once pressure stabilizes.
- Enter volume as 2.7 (for 2.0 (the volume of air in the syringe) + 0.7 (the volume of air in the pressure sensor) = 2.7 mL), then tap OK.

6. Collect More Data

- Repeat step 5 for:
    - 5 mL → Enter 5.7 mL
    - 10 mL → Enter 10.7 mL
    - 20 mL → Enter 20.7 mL
- Tap the Stop button when done.

7. Graph and Record

- Download and save a local copy of the **linked Excel spreadsheet**
- On the pressure vs. volume graph that appears, tap points to view data, then record them in Part A of the provided Excel sheet.

**Questions for Part A**

1. What happens to pressure when volume approximately doubles (5.7 → 10.7 mL)? Include values to support your answer.
2. What happens to pressure when volume approximately halves (20.7 → 10.7 mL)? What happens to pressure when volume approximately doubles (5.7 → 10.7 mL)? Include values to support your answer.
3. What happens to pressure when volume nearly quadruples (5.7 → 20.7 mL)? Include values to support your answer.
4. Is the pressure-volume relationship direct or inverse? Explain with your data. It may be useful to look at your graphs.
5. Predict the pressure at 40 mL. Show your calculation.

6. Predict the pressure at 1.0 mL. Show your calculation.
7. Which variables are constant in this experiment?
8. One way to determine if a relationship is inverse or direct is to find a proportionality constant, *k*, from the data. If this relationship is direct, *k* = P/V will all have quite similar answers. If it is inverse, *k* = P*V will have quite similar answers. On your Excel spreadsheet, create formulas to calculate the columns for P/V and P*V.
9. Calculate average and standard deviation for *k* calculated each way in Excel. Which *k* is more consistent: P/V or P*V?
10. Write an expression relating P, V, and a constant *k*. This is Boyle's law!
11. The values for k should be relatively constant. Calculate the average and standard deviation for each column using formulas in Excel. Which value (P*V or P/V) is closer to constant?
12. Plot the values pressure (y) against volume (x) by using the "Insert" menu on Excel and plotting a scatterplot. Add a linear trendline for this plot, making sure to display both the equation and $R^2$ value. Is it a good fit?
13. To confirm that an inverse relationship exists between pressure and volume, a graph of pressure as reciprocal of volume (1/volume) needs to be plotted. To do this in Excel:
    1. Highlight the cells B3:B9. Right click and select "Insert". Choose "Shift cells right".
    2. Add a new label "1/V" in cell B3, and create a formula in each cell for 1/V.
14. Plot the values pressure (y) against 1/V (x) by using the "Insert" menu on Excel and plotting a scatterplot. Add a linear trendline for this plot, making sure to display both the equation and $R^2$ value. Is it a good fit?
15. What is the relationship between the slope of this line (the "m" value) and the constant (*k*) you calculated?
16. Make sure both of these plots are clearly labeled and not overlapping in the spreadsheet. You may find it convenient move the chart to a new sheet. Click so the whole chart is selected, then right clicking to select "Move Chart" and select "New Sheet". Make sure to give each sheet an appropriate name like "Pressure vs. 1/V".

**PART B. Investigating Pressure /Temperature Relationship at Constant Volume and Moles**

**Data Collection**

1. *(If doing Part A then Part B, delete the current data by tapping on **File/New/Discard**.)* Connect the pressure sensor and add a temperature probe by connecting it into another CH port. You should now see two sensor readings on the screen (pressure and temp).
2. Go to the Meter screen with the top left meter icon, tap **Mode** in the upper right corner and change to **Events with Entr**y by tapping on the down arrow and making the selection. Tap on **OK**.
3. Change the pressure units to atmospheres by tapping on the box for pressure and selecting **atm**. Tap on the temp box and choose **Kelvin** for the units for temperature.
4. Set the temperature probe on the bench top and connect the pressure sensor to the small 50mL Erlenmeyer flask with the connector tubing and let it sit on the bench top as well. A stopper that comes loose results in the need to restart the entire experiment; so **make sure that the stopper is tightly seated in the flask**.
5. Start the experiment by tapping the green **start** button in the lower left corner. **You will not tap the stop button until you have done all four temperature tests**. Do not handle the tubing and flask any more than absolutely necessary since heat from your hands can change the flask's temperature. When the pressure reading stabilizes, *(do not tap the stop button)* tap the **keep** button and enter **1**.
6. Attach a utility clamp to the flask as demonstrated. Take the LabQuest with the pressure sensor connected to the Erhlenmeyer flask and the attached temperature probe to one of the water baths. One is an ice water bath, another is slightly above room temperature and the third is a very warm water bath. Submerge the flask and as much tubing as possible **without getting the pressure sensor wet**. Place the temperature probe into the bath as well. Gently rock the flask until the pressure reading on the LabQuest stabilizes. (This will take about 5 minutes.) The more patient you are the better your results will be.
7. After 5 minutes when the pressure reading stabilizes, tap the **keep** button and enter **2**.
8. Repeat step 6 and 7 until you have repeated the experiment in all three water baths entering **3** and **4** for each of these experiments after tapping the **keep** button. After the last data set has been "keep"ed, tap the **stop** button.
9. Tap the **Graph** tab and choose **Graph Options**… to open a window for setting up a proper display of the data. We need to determine if the pressure and temperature values are directly or inversely proportional. To start let's look at temperature on the x-axis and pressure on the y-axis. Choose **temperature** in the pull down menu for the **X-Axis Column** and then **check the Pressure under Run 1** and if necessary, **uncheck Temperature under Run 1**. Also, at the top of the window select "**Autoscale from 0**" and tap **OK**.

10. Tap **Graph** again, then **Show Graph**, and select **Graph 1**. To examine the data pairs on the displayed graph, tap the table icon to view your data. If you are completing part B first, download and save a local copy of the linked Excel spreadsheet. Under Part B, enter the pressure and temperature data you just took.

**Questions, Part B:**

17. If the temperature is increased, what does your data show happens to the pressure?

18. If the temperature is decreased, what does your data show happens to the pressure?

19. From your answers to the questions above and the shape of the curve in the plot of pressure vs temperature, do you think the relationship between the pressure and temperature of a confined gas is direct or inverse?

20. What experimental factors are assumed to be constant in this experiment? Question 4 on report sheet.

21. One way to determine if a relationship is inverse or direct is to find a proportionality constant, *k*, from the data. If this relationship is direct, *k* = P/T will all have quite similar answers. If it is inverse, *k* = P*T will have quite similar answers. On your Excel spreadsheet, create formulas to calculate the columns for P/T and P*T.

22. The values for k should be relatively constant. Calculate the average and standard deviation for each column using formulas in Excel. Which value (P*T or P/T) is closer to constant?

23. Write an expression relating P, T, and a constant *k*. This is Charles' law!

24. Using the average of your four estimates for a constant in the above table. Use this average to estimate the pressure of the gas in the flask if the temperature was 15°C. Show your work

25. Plot the values pressure (y) against temperature (x) by using the "Insert" menu on Excel and plotting a scatterplot. Add a linear trendline for this plot, making sure to display both the equation and $R^2$ value. Is it a good fit?

26. Using the equation from your graph, estimate absolute zero (in K and °C). It is the temperature where the pressure of your gas would be 0.00 atm. You will have to enter this value as your "y" and solve for "x". Show your work or do it in excel in a clearly labeled cell.

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# 11: Molar Mass of a Volatile Liquid

*Adapted from Kaya, J. J.; Campbell, J. A. Molecular Weights from Dumas Bulb Experiments. J. Chem. Educ. 1967, 44 (7), 394−395.*

**INTRODUCTION**

One of the properties that helps characterize a substance is its molar mass. If the substance in question is a volatile liquid, a common method to determine its molar mass involves using the ideal gas law, $PV = nRT$. Because the liquid is volatile, it can easily be converted to a gas. While the substance is in the gas phase, you can measure its volume, pressure and temperature. You can rearrange the ideal gas law to calculate the molar mass of the substance. (See Pre-Lab Assignment.) Also the gas constant that you will use in today's lab is R = 0.082058 $\frac{L*atm}{mol*K}$.

**PROCEDURE**

Obtain a round-bottom flask, a stopper, a chimney adaptor, a 600 mL beaker and a vial containing an unknown volatile liquid. Carefully measure the mass of the empty, dry round bottom flask and stopper. Prepare a boiling hot-water bath by heating about 575 mL of tap water in a 800 mL beaker (875 mL in a 1 Liter beaker) using a hot plate. Also add 7-10 boiling chips. Obtain a LabQuest and connect a temperature probe (units °C). Make sure the mode is time based and set for 180 seconds (3 minutes).

Pour all of your unknown volatile liquid into the round bottom flask and place the chimney adaptor on top. Suspend the round bottom flask using a clamp and stand as demonstrated by your instructor. When the water is fully boiling, carefully lower the round bottom flask into the boiling bath as far as possible. Top off any extra space in the beaker with hot tap water.

Maintain boiling as your sample of liquid vaporizes. Note that some of your sample will escape the round bottom flask through the chimney. This process also serves to flush the air out of flask. Monitor the liquid level in the flask as it evaporates. Careful observation of the chimney should reveal a plume of gas escaping. While the liquid is evaporating. record the atmospheric pressure in the lab using the barometer. Then prepare a cold water bath. In a 400 mL beaker, add cold tap water and ice to make 375 mL total This will be used to cool the flask in a later step.

Once all the liquid is vaporized, hold the temperature probe in the boiling water being careful not to touch the sides or bottom of the beaker or the round bottom flask and begin data collection on the LabQuest. Keep boiling the round bottom flask for three minutes *after* all of the liquid in the round bottom flask has vaporized. When the 180 seconds ends, replace the chimney with the stopper. Then lift the round bottom flask out of the boiling water bath using the clamp as a handle. Place the flask in the cold water bath prepared above. (The clamp may be removed from the flask when convenient.) Every fifteen seconds or so, it is important to partially lift the stopper for a second to allow air to flow into the flask. Continue this process as you cool the flask for a total of 3 minutes.

Examine your temperature graph and record the highest temperature of the boiling-water bath, which will be used in the ideal gas law calculations. Now remove the flask from the cool water and carefully dry it completely. Partially remove the stopper one last time to equalize the pressure then measure the mass of the flask, stopper and condensed unknown liquid.

**Determining total volume of flask with stopper inserted**

Pour condensed liquid from your round bottom flask into the organic waste container in the hood. Rinse the round bottom flask with a few mLs of acetone and then with some DI water. Now fill the flask to the top with DI water. Insert the stopper carefully squeezing out water and not allowing air bubbles to form. Dry the flask and stopper completely. It would be ideal to obtain the mass of the entire flask at this point. But our balances can only weigh 200 grams at a time. Therefore you need to weigh the water that the flask now contains in several steps to get the total mass of the water in the flask. So you need to determine a procedure that obtains the total mass of water in the round bottom flask. Then convert the resultant mass of water into milliliters to calculate the total volume of the flask. Download a local copy of the linked Excel spreadsheet to enter your data. Return the round bottom flask, stopper, chimney and 600 mL beaker to the cart. ***Answer at least the first 3 questions on the next page before leaving lab today.***

1. What was your procedure for determining the mass of water that filled the flask, and how did you use that to determine the volume of the flask? Include all data obtained and calculations needed for this determination.

2. Your unknown liquid is one of the following: hexane ($C_6H_{14}$), isopropanol ($C_3H_8O$), 2-butanone ($C_4H_8O$), ethanol ($C_2H_6O$), or 1,1,1-trichloroethane ($C_2H_3Cl_3$).

Identity of the unknown liquid ________________

Show how you identified your liquid.

3. After identifying your unknown, open the linked Excel spreadsheet, switch the sheet to “Class Data” and enter your data.

4. If all the vapor in your flask had not condensed to a liquid when you cooled the flask, how would your calculations have been affected?

5. How would your experiment have been affected if you had used a different initial amount of the unknown compound?

6. Why does the stopper have to be partially removed so frequently as the flask cools?

7. Once there are at least 6 data points entered into the spreadsheet, copy the data into your local copy.

8. On your local copy, click in cell C3 (literature molar mass, first entry). Click the “Data” menu, then ”sort”. Choose “Sort by” then select “Literature” and click “OK”.

9. Now all the data for the same unknowns is together. Find the average **% error** for each unknown and the standard deviation of the **% error** for any unknown with three or more entries. Make sure these values are calculated in Excel and clearly labeled.

10. Compare the accuracy of the different unknowns. Which molecule was most accurate? Least accurate? Did the unknowns dominated by hydrogen bonding vs. dipoles vs London forces seem to exhibit any difference in accuracy?

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# 12: Solar Energy Storage

*Adapted from ChemTrek Chapter 6, Thompson S, Prentice Hall Publishers, 1989.*

**Section A. Calibration of a Calorimeter**

The heat capacity measurement is accomplished by using a cooling curve analysis of an experiment involving the addition of a known volume of hot water to a known volume of room temperature water in the microcalorimeter. The temperature changes can be monitored by a LabQuest and temperature probe. The heat given by the hot water will need to heat both the cold water and the calorimeter, and so can be used to find the heat capacity of the calorimeter you created:

$$-q_{released\ by\ hot\ water} = q_{absorbed\ by\ system} = q_{water} + q_{calorimeter}$$

$$-\underset{\text{(hot water)}}{specific\ heat * mass * \Delta T} = \underset{\text{(cold water)}}{specific\ heat * mass * \Delta T} + heat\ capacity\ calorimeter * \Delta T$$

1. Insulate a 50 ml beaker by wrapping it with a paper towel and tucking it into a 100 ml beaker. The 50 mL beaker should be easily removed and replaced without disturbing the paper towel insulation. Turn on the LabQuest and connect the temperature probe to it. Change the time duration for the experiment to 10 minutes (600 seconds). Change the temperature range of the graph to display from 20 °C to 50 °C.

2. Measure 5.0 mL of DI water with a graduated cylinder and place in a hot bath. Take note of the bath temperature using the LabQuest temperature probe and record it in a local copy of the linked Excel spreadsheet as the initial temperature of the *hot* water.

3. Transfer a 15.0 mL DI water (measured with a 50 mL graduated cylinder) to the calorimeter. Start the run by hitting “play”. After 5 minutes, add all of the hot water to the hot water bath. Pipette 3 times up and down with a beral pipette to mix. Continue the experiment until the temperature after mixing has stabilized for one minute.

4. After stopping the experiment, pour out all water and rinse twice with room temperature DI water. Dry the container and refill with 15.0 mL of DI water.

5. Your graph should look similar to the graph below. Estimate the initial temperature of the room temperature water, $T_i$, from the initial 5 minutes of the graph, extending any line to half-way through the temperature increase. Estimate the final temperature from the final portion of the graph, extended backward to the same point. Record those temperatures in your local copy of the linked Excel spreadsheet.

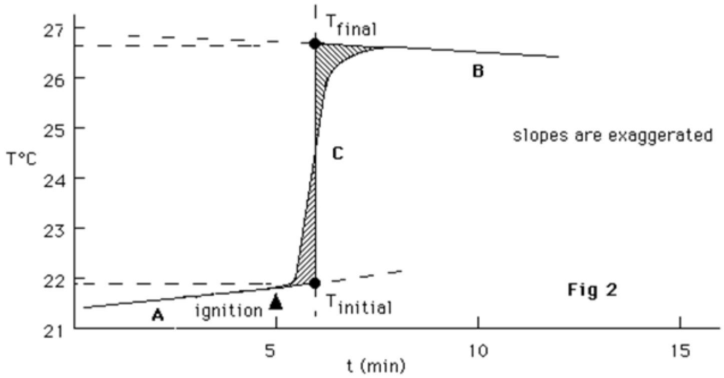


Fig 2

6. Calculate the heat lost by the hot water using the density and specific heat given in the spreadsheet. Calculate the heat gained by the cold water in the same way. The difference between these is the heat gained by the calorimeter. The temperature change for the calorimeter is the same as the temperature change for the room temperature water. Use the heat gained by the calorimeter and the temperature change to solve for the heat capacity in units of J / °C.

**Section B. Determination of the Specific Heat of Granite Rock Used for Solar Energy Storage**

1. Begin a new ten-minute run with the temperature sensor in the water you added after rinsing in the last experiment. Make sure you have 1 minute of stable data.

2. Go to the oven and record the temperature shown on the thermometer. Record this on the “Granite” sheet as the initial temperature for the granite. Perform the calculations for this section here.

3. Quickly remove 3 rocks from the oven and place them directly into your calorimeter. Please make sure to use gloves or tongs.

4. Pipette 3 times up and down with a beral pipette to mix. Continue the experiment until the temperature after mixing has stabilized for one minute.

5. After stopping the experiment, carefully pour off the water and remove the rocks. Dry them with a paper towel.

6. Rinse the calorimeter twice with room temperature DI water. Dry the container and refill with 15.0 mL of DI water.

7. When the rocks are dry, weigh them and record the weight.

8. Now the heat balance is the heat lost by the rocks is equal to the heat gained by the room temperature water and calorimeter. Use the heat capacity of the calorimeter you calculated in Section A to find the heat lost by the rocks, then find the heat capacity of the rocks using the heat lost, mass, and temperature change.

**Section C. Determination of the Heat of Crystallization of Sodium Thiosulfate Pentahydrate, $Na_2S_2O_3$•$5H_2O$**

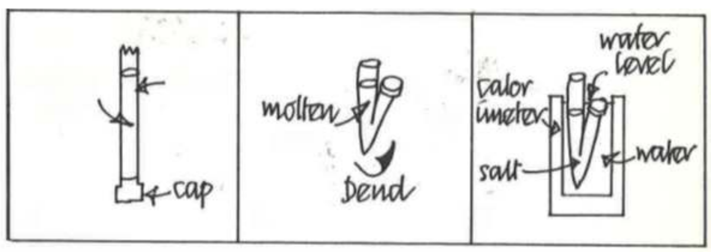
1. Cut 5 cm off the end of a straw and discard the 5 cm piece. You now have a piece about 15 cm long.

2. Insert a plastic cap *tightly* into one end of the 15 cm piece.

3. Weigh the capped straw and record the weight.

4. Make a mark on the straw 6 cm from the cap.

5. Using a thin-stem pipet, transfer liquid salt from the bath of molten sodium thiosulfate pentahydrate into the straw until it reaches the 6 cm mark. Set aside for a few minutes to allow the salt to cool to the crystallizing temperature (~48 °C). This is the initial temperature for the salt. Record it on your local copy of the linked Excel spreadsheet.




6. Begin a new ten-minute run with the temperature sensor in the water you added after rinsing in the last experiment. Make sure you have 1 minute of stable data.

7. Check on the salt. It should still be liquid.

8. Quickly add a seed crystal and make sure that the salt begins to crystallize.

9. Bend the straw at the center of the liquid salt and place doubled up into the water in the calorimeter, ensuring that the molten salt is completely covered, as shown above.

10. Pipette up and down slowly with a beral pipette to mix. Continue pipetting gently up and down to mix – the heat exchange is slower. Continue the experiment until the temperature after mixing has stabilized for one minute.

11. After stopping the experiment, remove the bent straw from the water and dry it with a paper towel.

12. Record the weight of the straw + crystallized salt.

13. Now the heat balance is the heat lost by the salt is equal to the heat gained by the room temperature water and calorimeter. Use the heat capacity of the calorimeter you calculated in Section A to find the heat lost by the salt, then find the heat capacity of the rocks using the heat lost, mass, and temperature change.

**Post-Lab Questions:**

1. Do you think that a classical mercury thermometer could have been used to carry out the microcalorimeter experiments? Give reasons for your answer.

2. Do you think that a heat-insulating top for the microcalorimeter would have improved the results?

3. Straws happen to be excellent containers for the determination of the heat of crystallization of molten salts. Why?

4. Calculate and compare the amount of heat energy stored by 1 pound of water, 1 pound of granite, and 1 pound of sodium thiosulfate pentahydrate. Assume a $\Delta T$ of 40 ° C for the temperature change of the water and the rock.

5. If you could not afford to pay elves (or students) to add seed crystals at the appropriate time to the solar energy storage tank to make the molten salt crystallize, how could you insure crystallization?

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# INDEX

# GLOSSARY

**Sample Word 1** | Sample Definition 1

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